WOODWARD HRT

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VALIDATION REPORT

FOR

COMPUTER DATA ACQUISTION AND MONITOR SYSTEM REPLACING THE X&Y RECORDER ON FLOW GRIND TEST STANDS

Prepared for

Woodward HRT

Prepared by: Dave Bailey **Date:** 13 August 2012

FOR APPROVALS, SEE ATTACHED DOCUMENT RELEASE REPORT (DRR)

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1.0 INTRODUCTION

1.1 Scope

This Woodward HRT (WHRT) procedure describes the method of comparing the current plotting method of using X & Y Recorders to a new method of using a Computer Data Acquisition (DAQ) and a Monitor System for the Flow Grind Test Stands. The test stands still have all the same hydraulics, cylinders, transducers, and electronics.

1.2 Purpose

This validation report is to verify that the Computer DAQ and Monitor System can perform as well as, if not better than, the current X & Y Recorder for the flow grind test stands.

1.3 Acronyms/Abbreviations

ATP	Acceptance Test Procedure
DAQ	Data Acquisition
WHRT	Woodward HRT

2.0 APPLICABLE DOCUMENTS

The following documents form a part of this report to the extent specified herein. Unless otherwise specified, the latest issue of the applicable documents specified shall be used. In the event of a conflict between the documents referenced herein and the contents of this report, the contents of this report shall take precedence.

2.1 Industry Documents

ANSI/NCSL-Z540.3	Requirements for the Calibration of Measuring and Test Equipment

2.2 Woodward HRT Documents

3-QCI-00181	Calibration System Procedure

HRQP 7.001	Contamination Control Procedure
P05-20 Rev O-2	Product Acceptance Software

3.0 GENERAL REQUIREMENTS

3.1 Overview

Keep in mind, these X & Y Recorders have done their job throughout the years but are no longer manufactured, and is becoming difficult to find repair parts for them. Even with the new DAQ system, there is still operator sensitivity in the plotting process. As continued improvement in software and equipment upgrades are made, additional steps will be taken to reduce variation from operator to operator.

3.2 DAQ Setup Information

A total of three parts were run on test stand 4-2. Each part number was run five times on the current X & Y Recorder method and five times using the new Computer DAQ and Monitor System method.

Three main feature specifications were measured in the ranges shown below:

NOTE: No-load Drift is not a primary feature. It is used mainly as indicator information to the operator. It indicates the relationship of the four metering edges to each other. Also, it informs the operator which edge to grind to adjust for neutral pressure.

Neutral pressure is measured by visually looking at the two cylinder gages on the test stand.

a. Lap condition from 0.0001 underlap to 0.0004 over lap.

b. Pressure Gain from 3500 psi to 7000 psi.

c. Leakage from 0.030 to 0.300 gpm.

3.3 Flow Grind Engineering Specifications

a. P/N 28007168-008, 22282790-104:

1. Supply pressure: 3000 psi
2. Overlap: 0.000300 to 0.000400
3. Pressure gain: 4600 psi/0.001 in. minimum
4. Pressure gain at 1200 psi: 0.000522 inch maximum (when plotted)
5. Leakage: 0.3000 gpm maximum
6. No-load Drift (ref): Not required, was plotted for data

b. P/N 28009498-102, 22284250-104:

1. Supply pressure: 3000 psi
2. Overlap: 0.000400 to 0.000800
3. Pressure gain: 3500 psi/0.001 in. minimum
4. Pressure gain at 1200 psi: 0.000685 inch maximum (when plotted)
5. Leakage: 0.0900 gpm maximum
6. No-load Drift: 2900 psi maximum (at lowest psi near null)

c. P/N 28000747-011, 22253726:

1. Supply pressure: 3000 psi
2. Overlap: 0.000200
3. Underlap: 0.000100
4. Pressure gain at 1200 psi: 0.003280 inch maximum (when plotted)
5. Leakage: 0.0300 gpm maximum
6. No-load Drift (ref): Not required, was plotted for data

4.0 DATA RESULTS

4.1 DAQ Results

4.1.1 Summary of Data Results by Part Number

On average, the standard deviation for P/N 22282790-104 was approximately two times smaller (tighter spread) on the new Computer DAQ and Monitor System than on the current X & Y Recorder. This high variation reduction shows that the new Computer DAQ and Monitor System would have better repeatability than the current X & Y Recorder.

On average, the standard deviation for P/N 22284250-104 was about 20% smaller (tighter spread) on the new Computer DAQ and Monitor System than on the current X & Y Recorder. The best improvement was seen in measuring No-load Drift, where the standard deviation was cut nearly in half between the two systems (24.6 vs. 45.7 psi).

On average, the standard deviation for P/N 22253726 was about 11% smaller (tighter spread) on the new Computer DAQ and Monitor System than on the current X & Y Recorder. The data matched up nicely for this part.

4.1.2 Test for Equal Variances Data

Plots are shown in Figures 1 through 28, which gives a summary of the variance between the manual (X&Y Recorder) and digital (computer DAQ) samples of data for each part number.

For P/N 22282790-104, in every case the variance of the digital plotter was less than the variance of the current manual pen plotter method. A bias for shifting the mean to a lower value in the case of the digital plotter was seen across the five different graphs, but this shift is negligible with respect to the tolerance bands of the measured criteria, and a similar shift is not found within the other two part numbers.

For P/N 22284250-104, the variance of the digital plotter, when measuring Lap Condition at 0.001 and 0.002, was slightly higher than the variance of the current manual method. This is a negligible find, and could simply be a result of operator sensitivity during plotting. Variance in measurements of Leakage, No-load Drift, and Pressure Gain was shown to be lower for the digital plotter when compared to the current manual plot method.

For P/N 22253726, the variance between the digital and manual plot methods is small. Overall, the figures show that the variance of the digital plotter is better than, if not the same as, the current manual method. It should be noted that the variance for the digital plotter when measuring Pressure Gain is less than the variance of the current Manual plotter (across all part numbers).

4.1.3 Conclusion

The variation is less for the DAQ system. Overall the curves are narrower than the curves for the X & Y Recorder.

All data and calculations together support that the new DAQ system is capable of measuring just as well as, if not better than, the current X & Y Recorder.

4.2 Flow Grind Results

The results are shown in Tables 1 through 3.

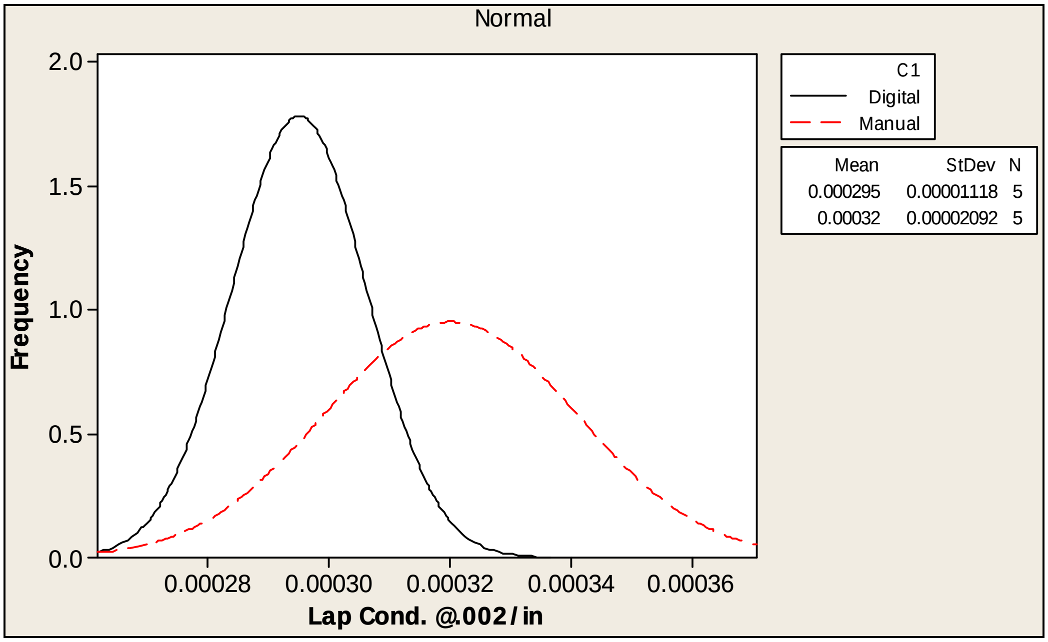


Figure 1. P/Ns 28007168-008, 22282790-104, S/N 777 – Lap Condition at 0.002/in.

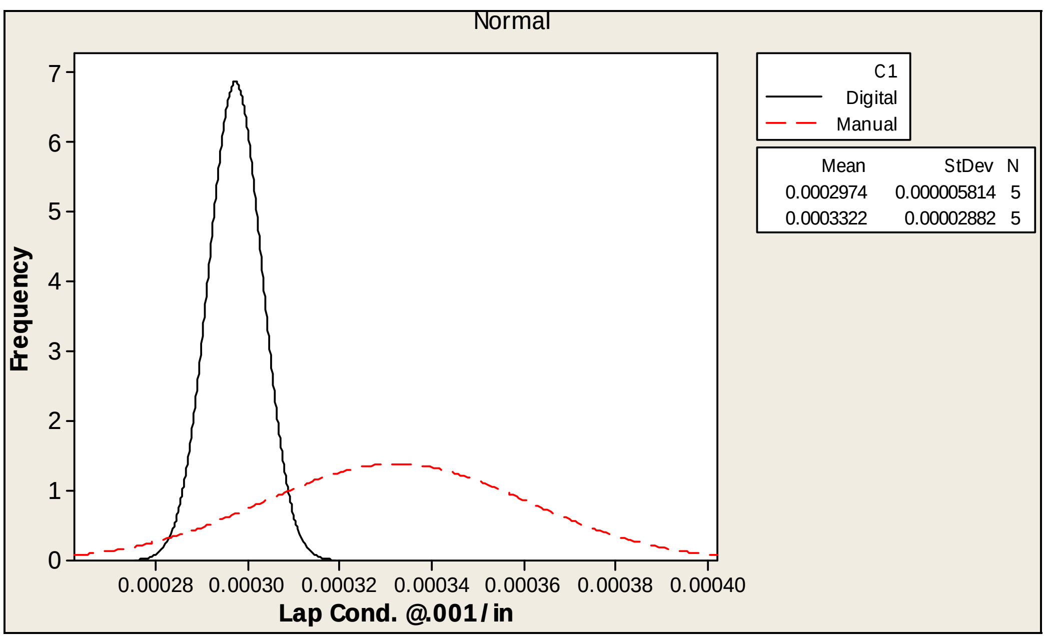


Figure 2. P/Ns 28007168-008, 22282790-104, S/N 777 – Lap Condition at 0.001/in.

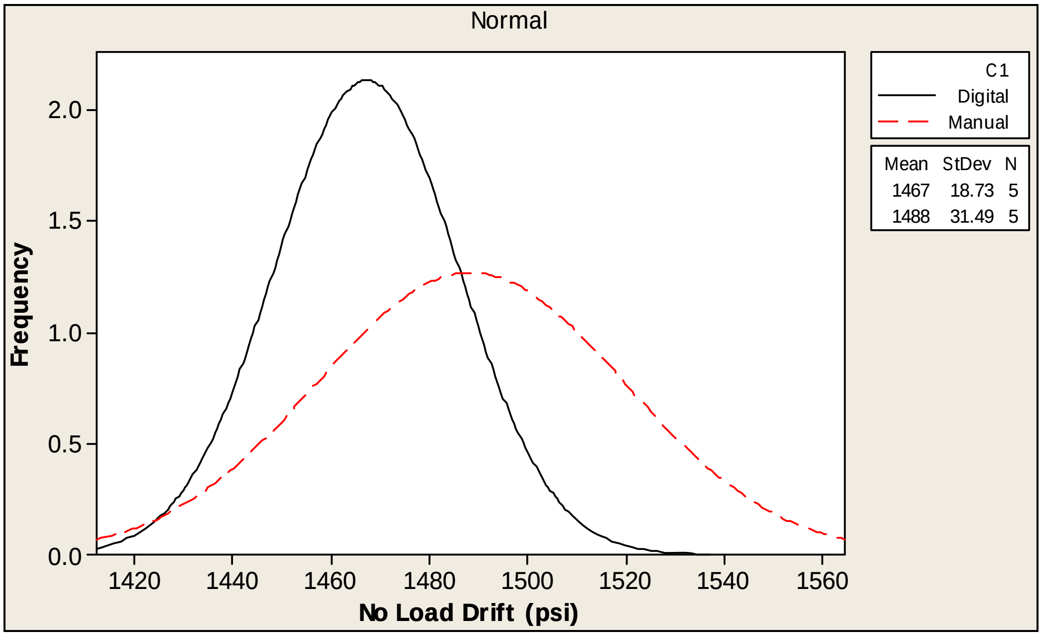


Figure 3. P/Ns 28007168-008, 22282790-104, S/N 777 – No-load Drift

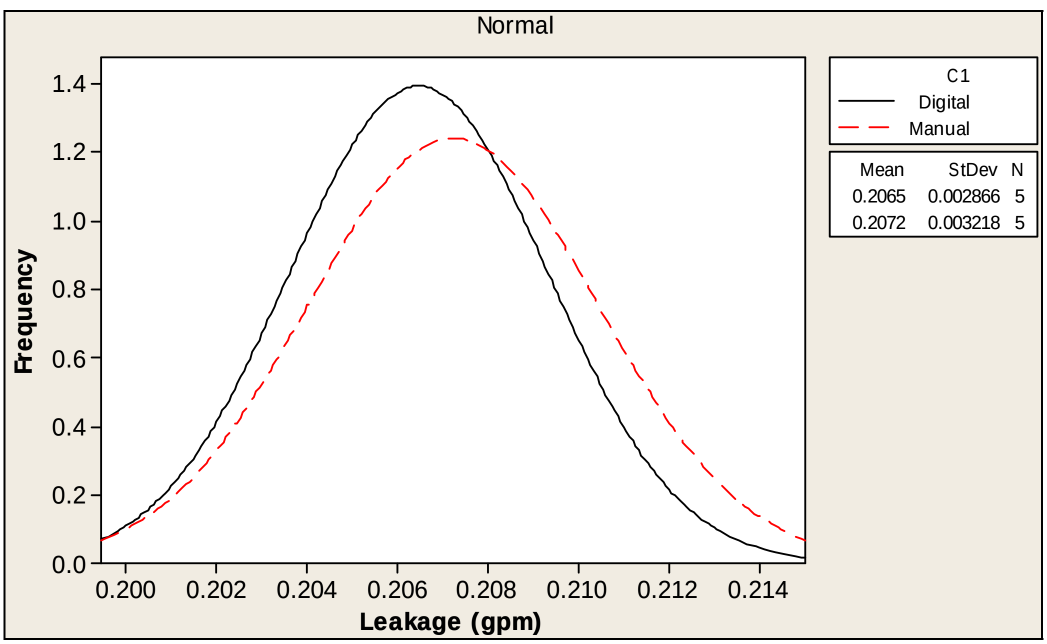


Figure 4. P/Ns 28007168-008, 22282790-104, S/N 777 – Leakage

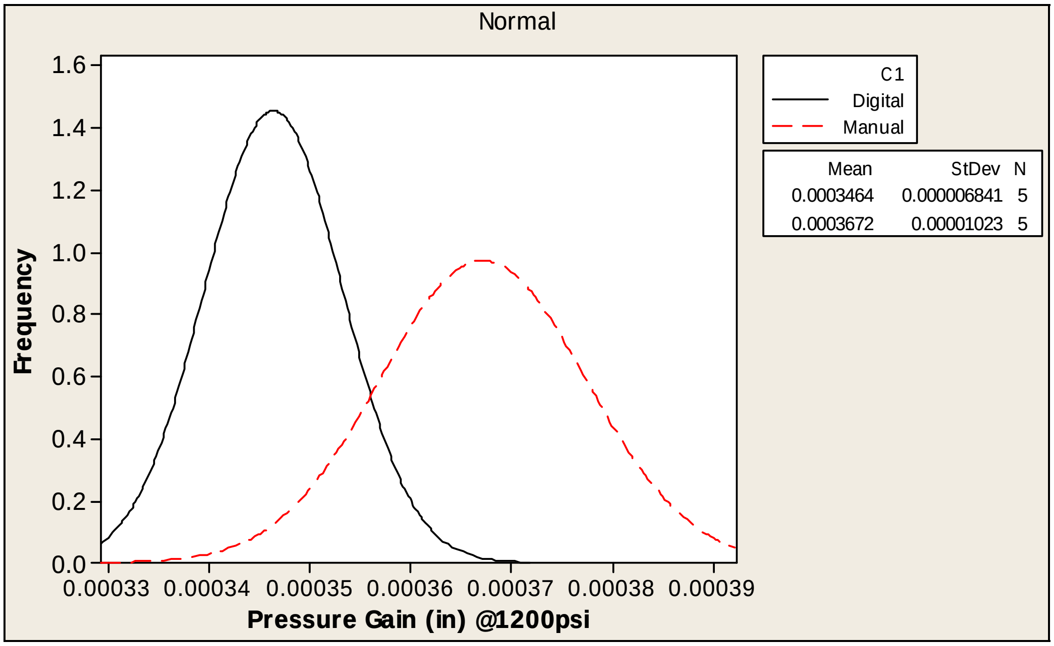


Figure 5. P/Ns 28007168-008, 22282790-104, S/N 777 – Pressure Gain at 1200 psi

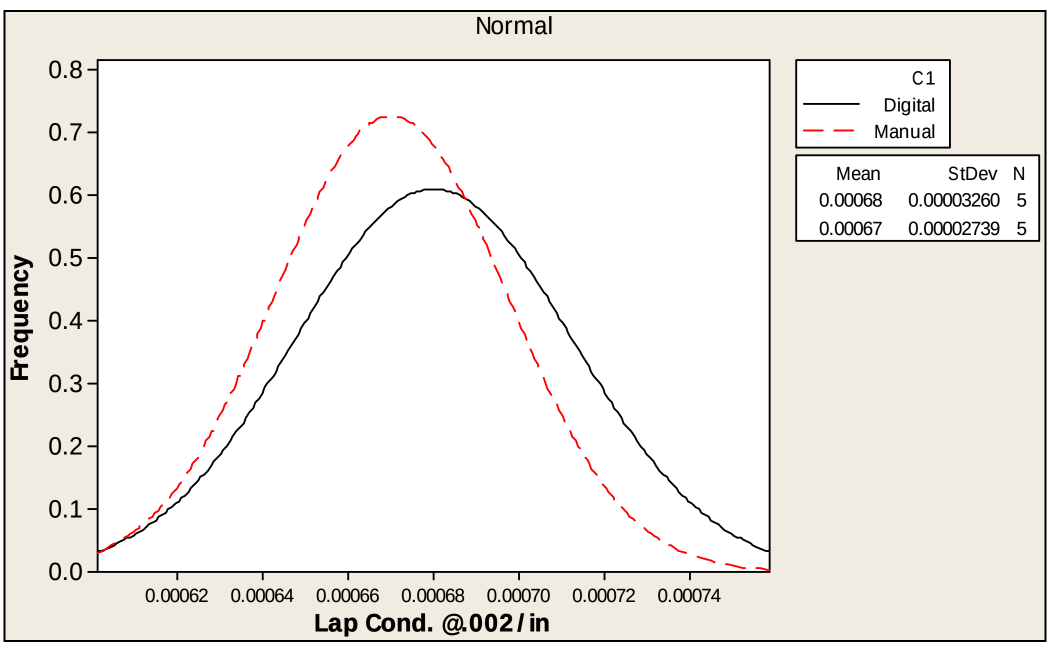


Figure 6. P/Ns 28009498-103, 22284250-104, S/N 365 – Lap Condition at 0.002/in.

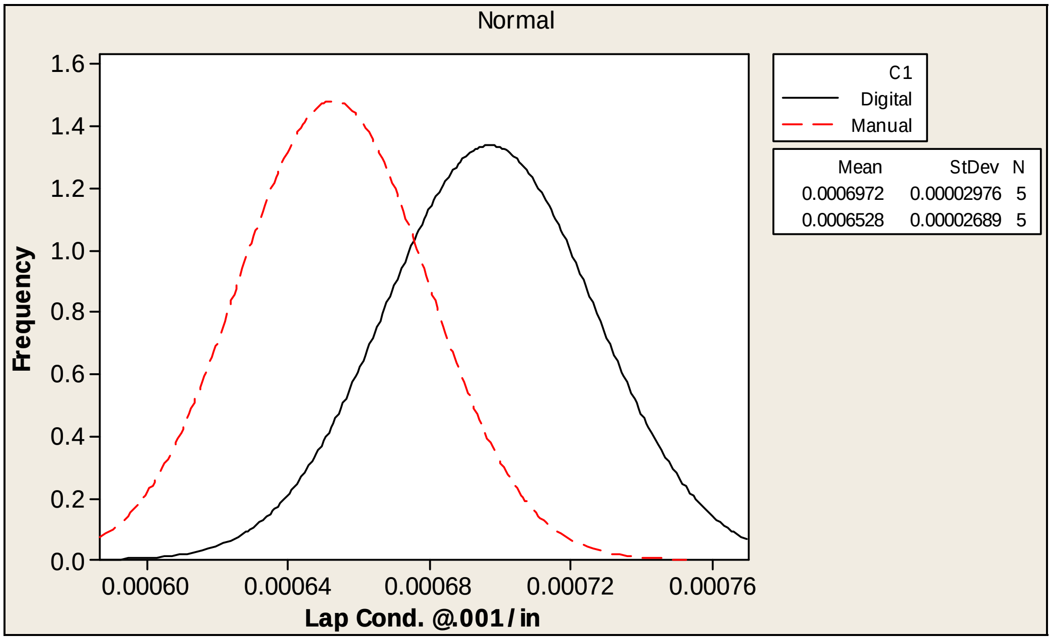


Figure 7. P/Ns 28009498-103, 22284250-104, S/N 365 – Lap Condition at 0.001/in.

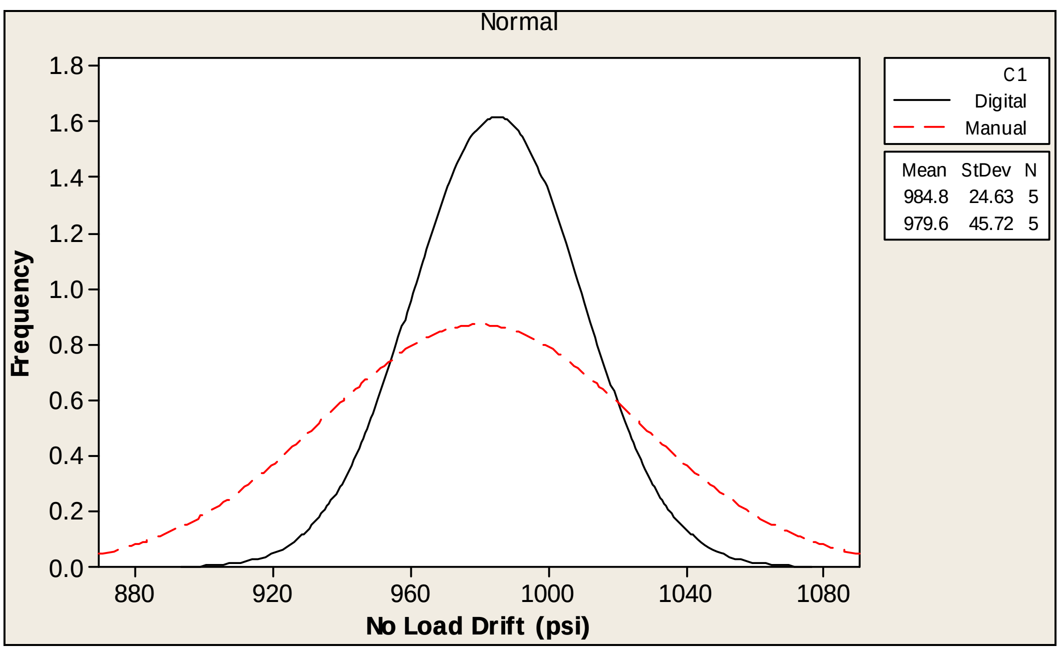


Figure 8. P/Ns 28009498-103, 22284250-104, S/N 365 – No-load Drift

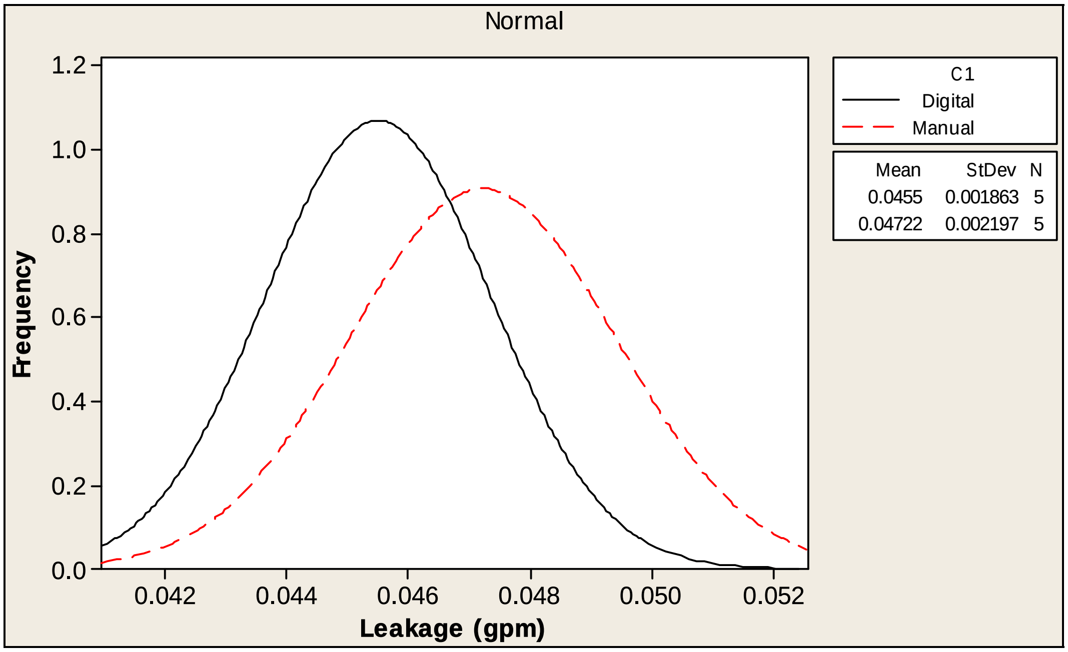


Figure 9. P/Ns 28009498-103, 22284250-104, S/N 365 – Leakage

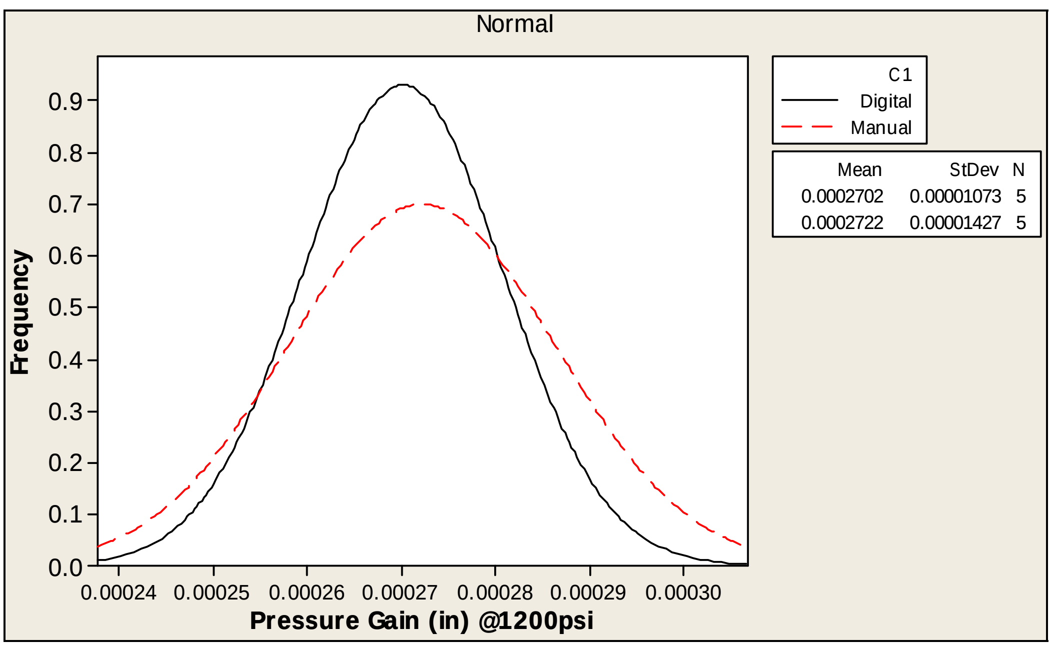


Figure 10. P/Ns 28009498-103, 22284250-104, S/N 365 – Pressure Gain at 1200 psi

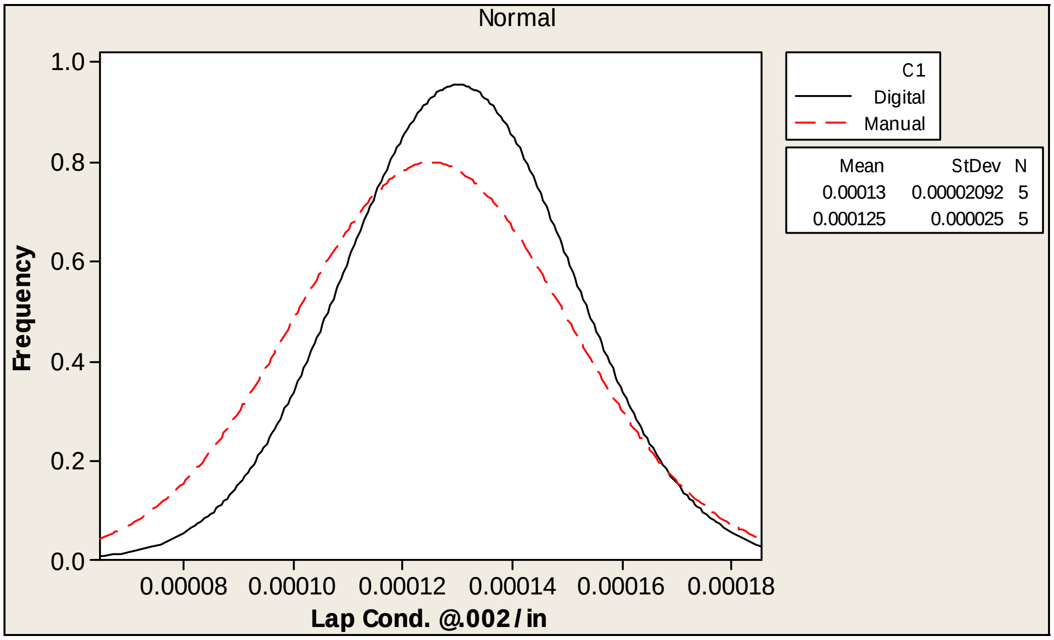


Figure 11. P/Ns 28000747-011, 22253726-104, S/N 835 – Lap Condition at 0.002/in.

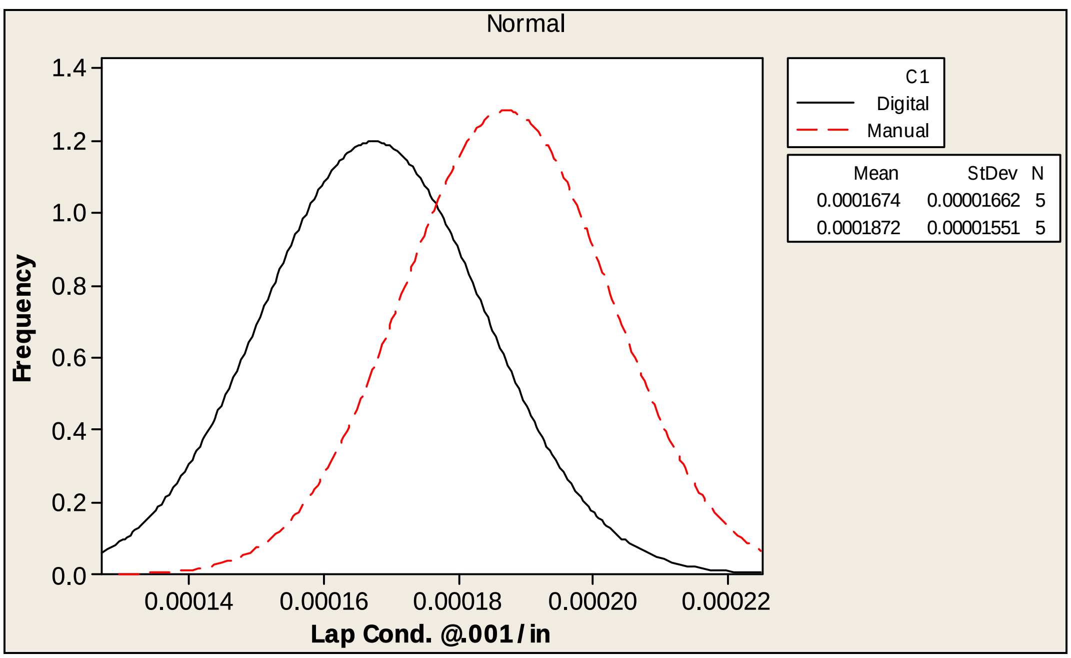


Figure 12. P/Ns 28000747-011, 22253726-104, S/N 835 – Lap Condition at 0.001/in.

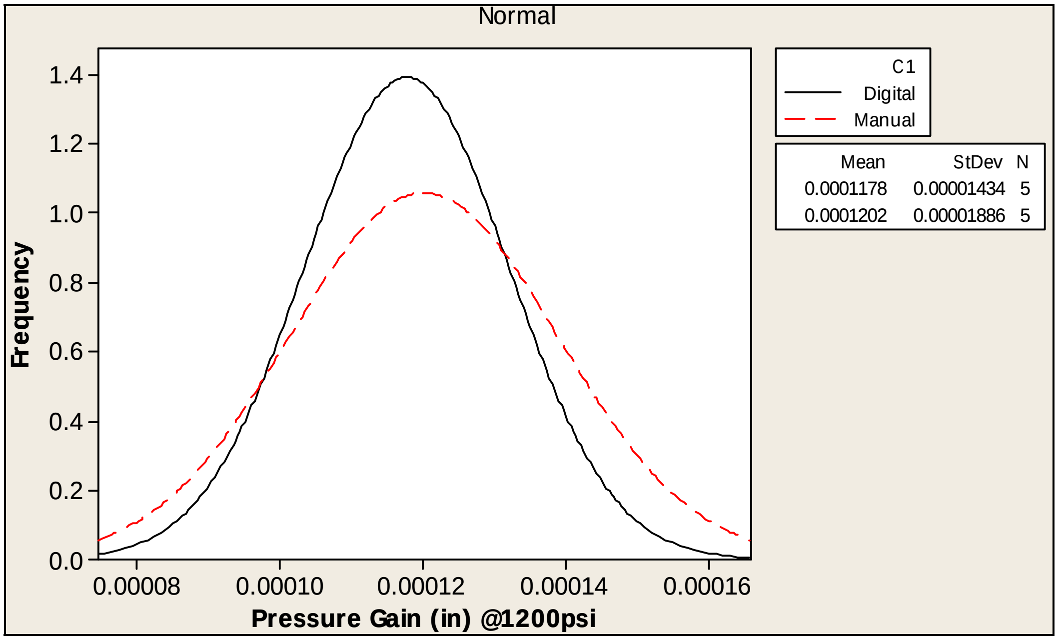


Figure 13. P/Ns 28000747-011, 22253726-104, S/N 835 – Pressure Gain at 1200 psi

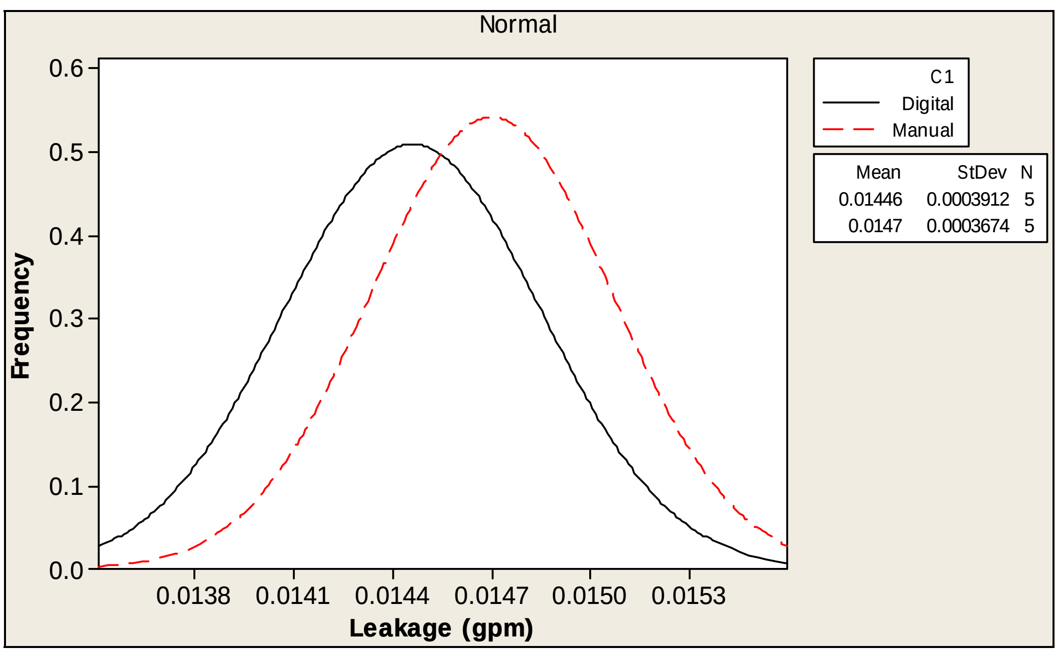


Figure 14. P/Ns 28000747-011, 22253726-104, S/N 835 – Leakage

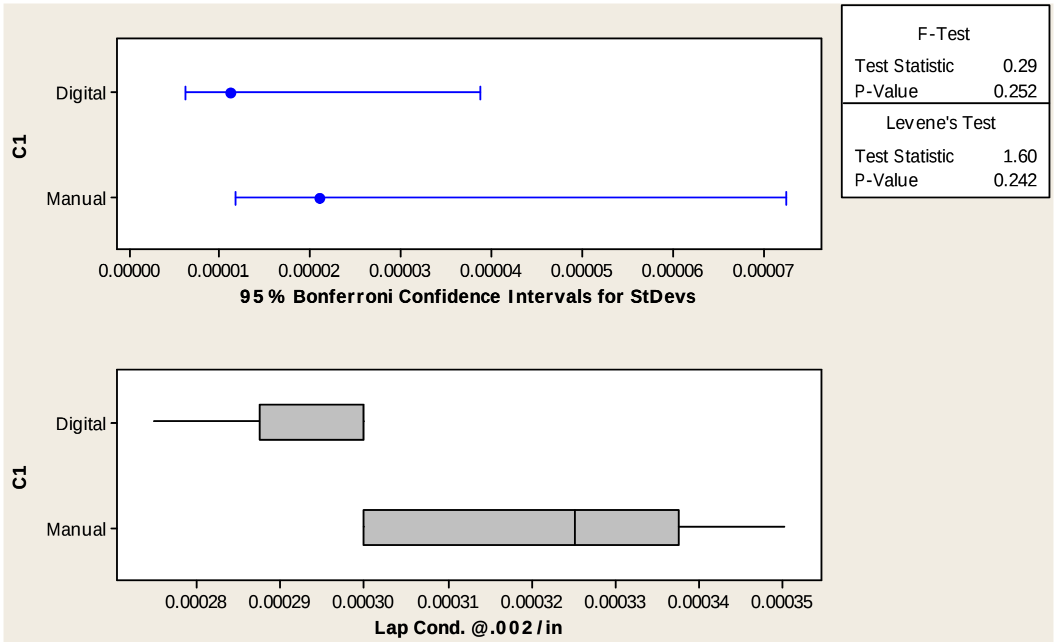


Figure 15. P/N 22282790-104 Test for Equal Variances – Lap Condition at 0.002/in.

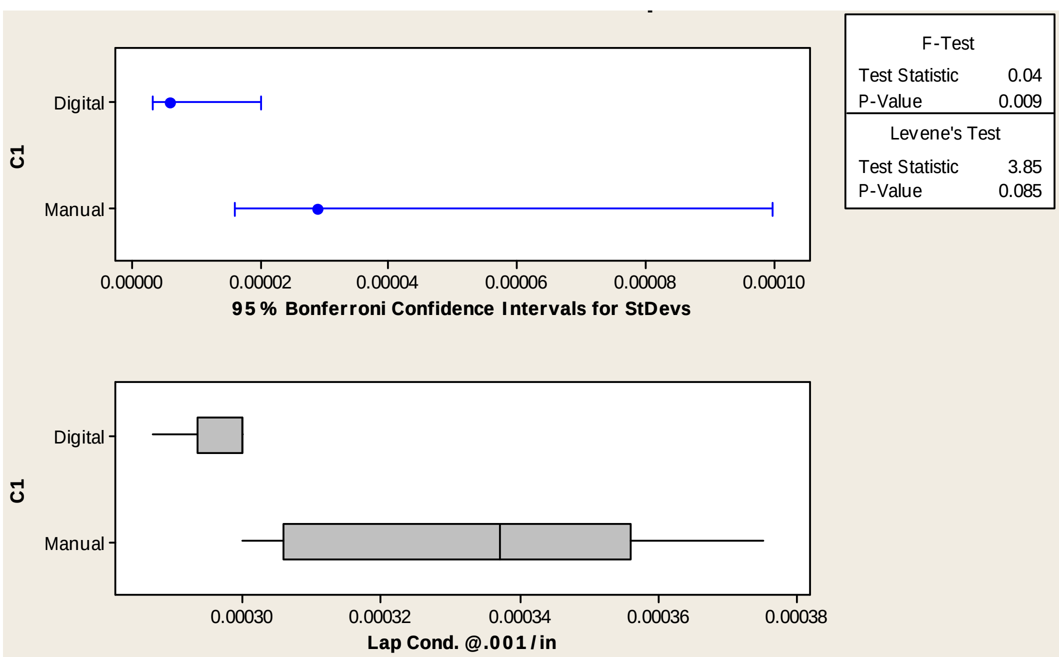


Figure 16. P/N 22282790-104 Test for Equal Variances – Lap Condition at 0.001/in.

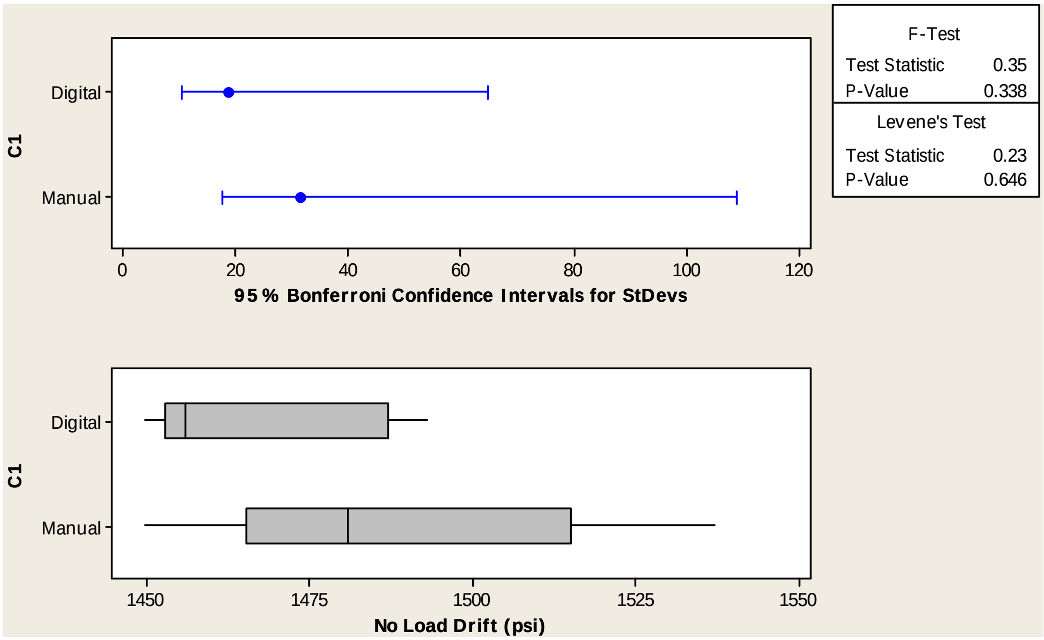


Figure 17. P/N 22282790-104 Test for Equal Variances – No-load Drift

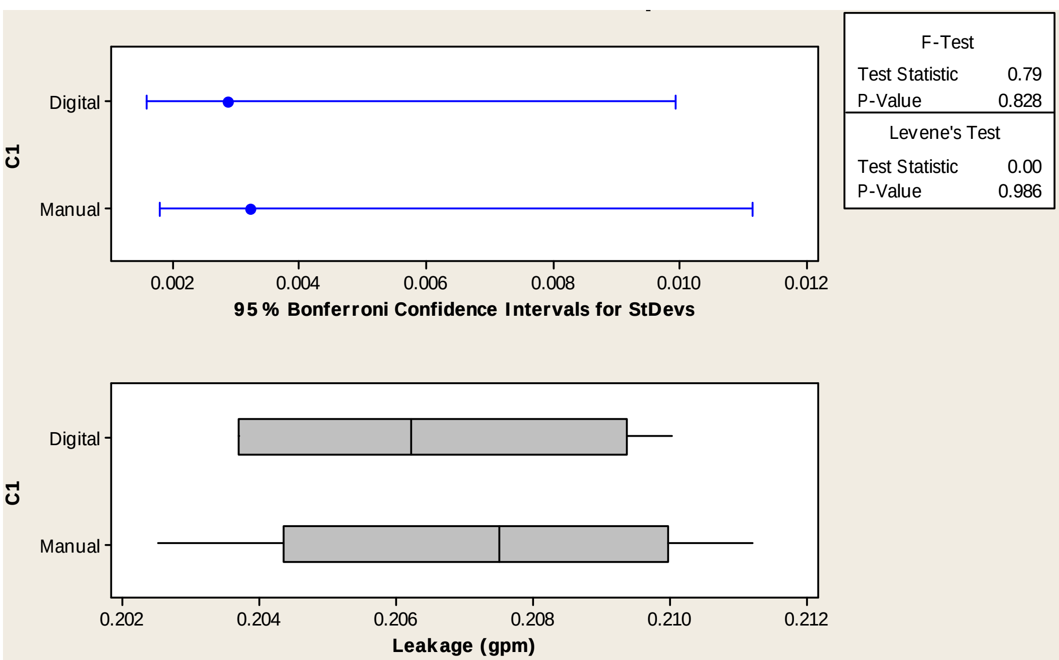


Figure 18. P/N 22282790-104 Test for Equal Variances – Leakage

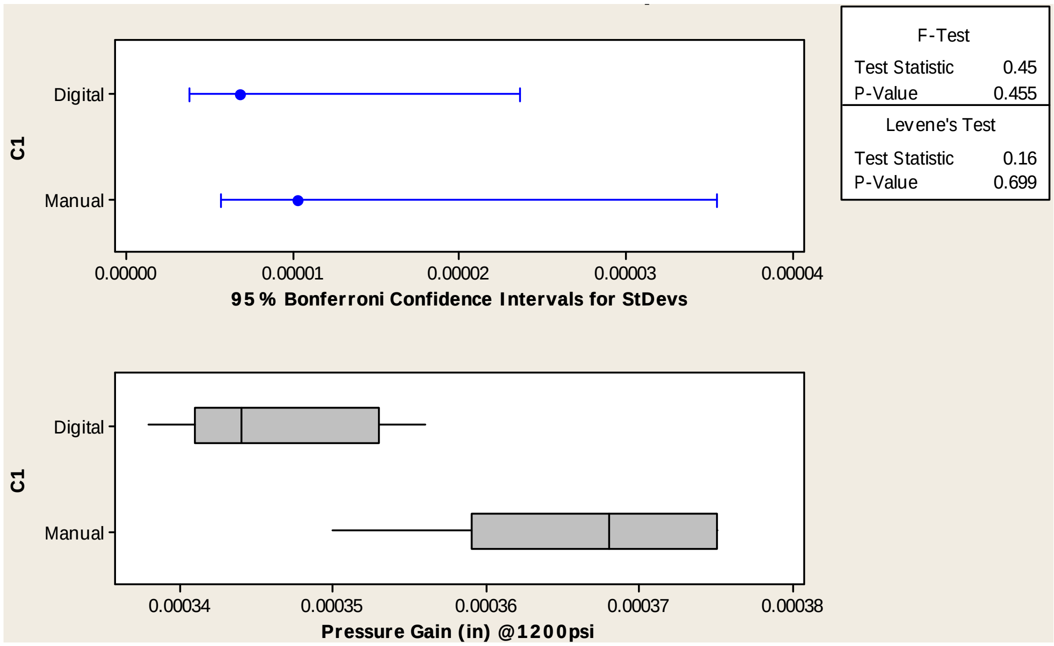


Figure 19. P/N 22282790-104 Test for Equal Variances – Pressure Gain at 1200 psi

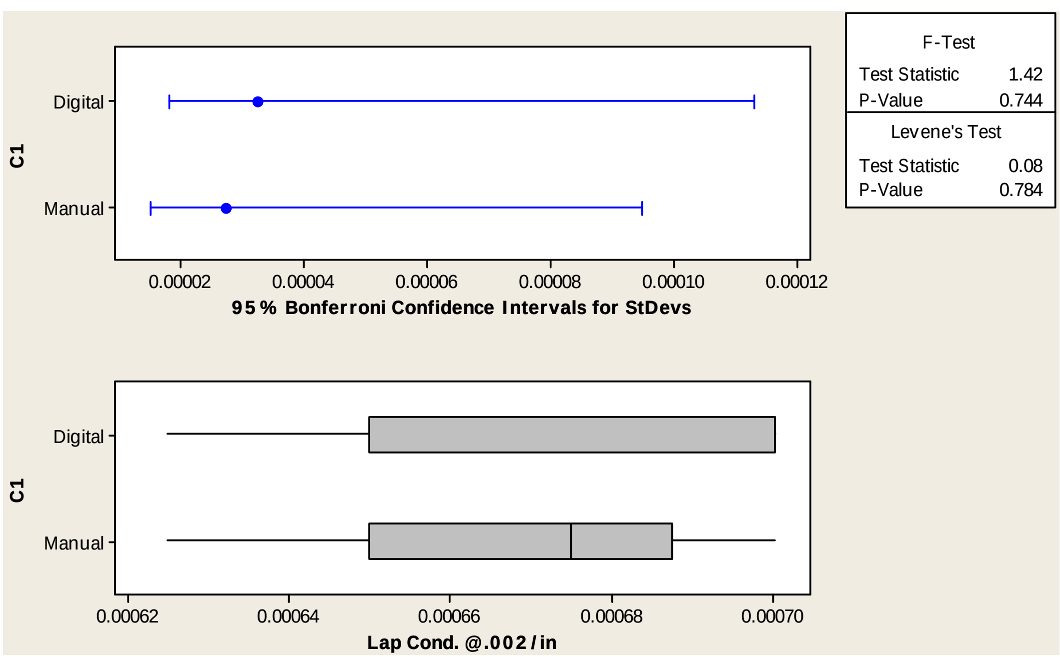


Figure 20. P/N 22284250-104 Test for Equal Variances – Lap Condition at 0.002/in.

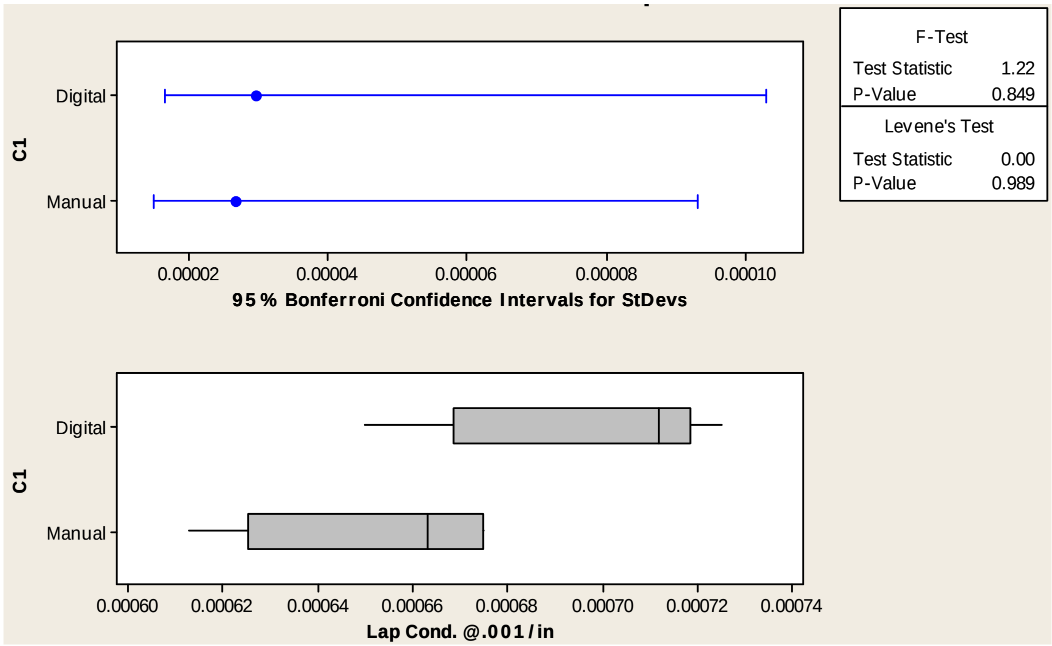


Figure 21. P/N 22284250-104 Test for Equal Variances – Lap Condition at 0.001/in.

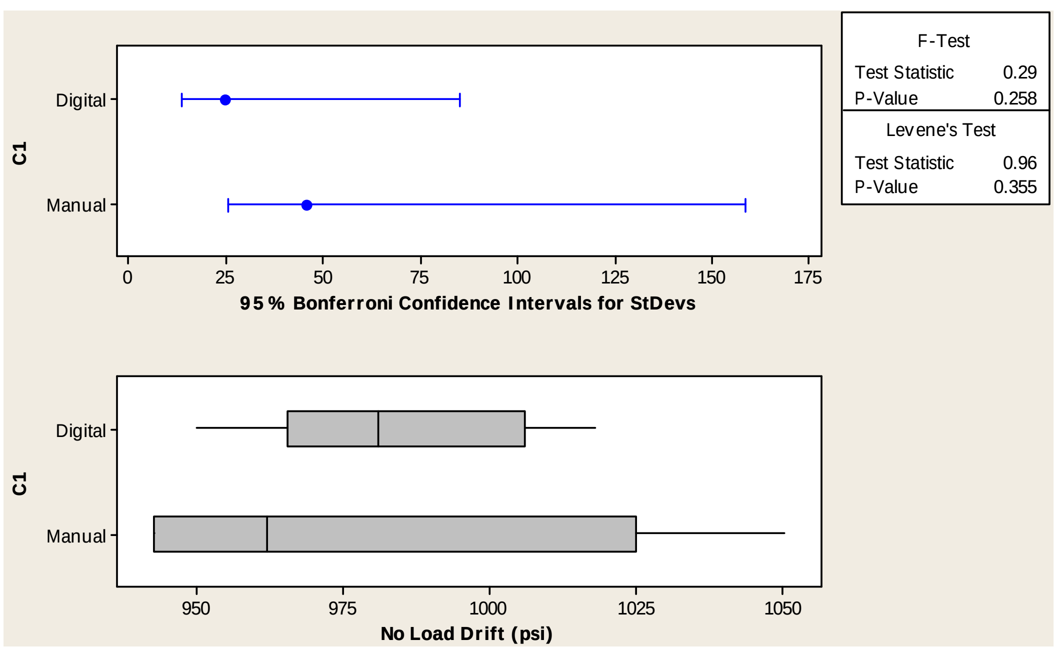


Figure 22. P/N 22284250-104 Test for Equal Variances – No-load Drift

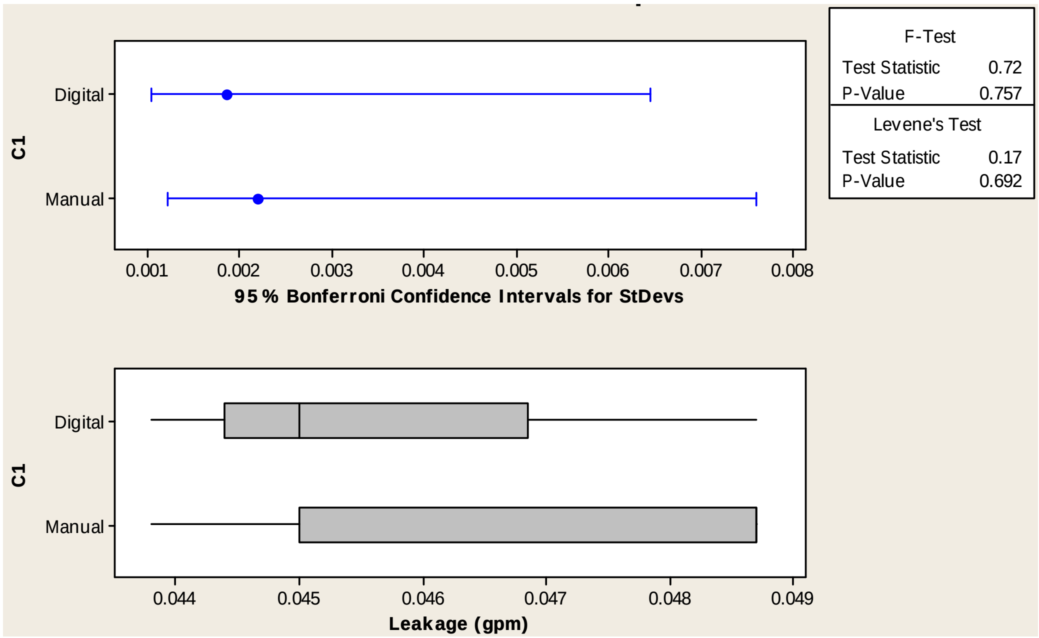


Figure 23. P/N 22284250-104 Test for Equal Variances – Leakage

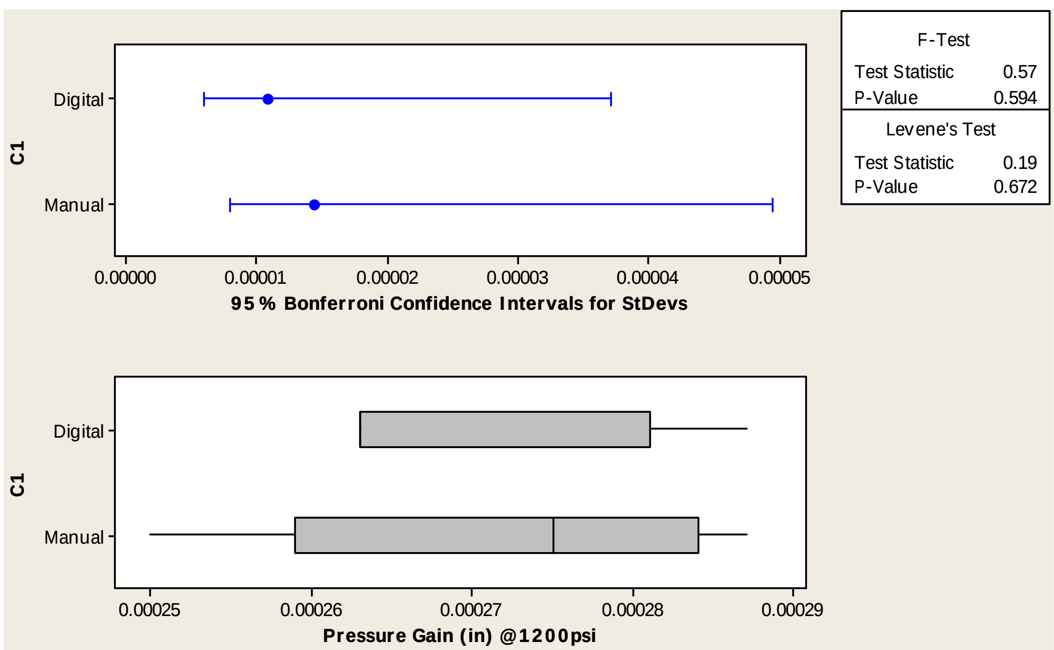


Figure 24. P/N 22284250-104 Test for Equal Variances – Pressure Gain at 1200 psi

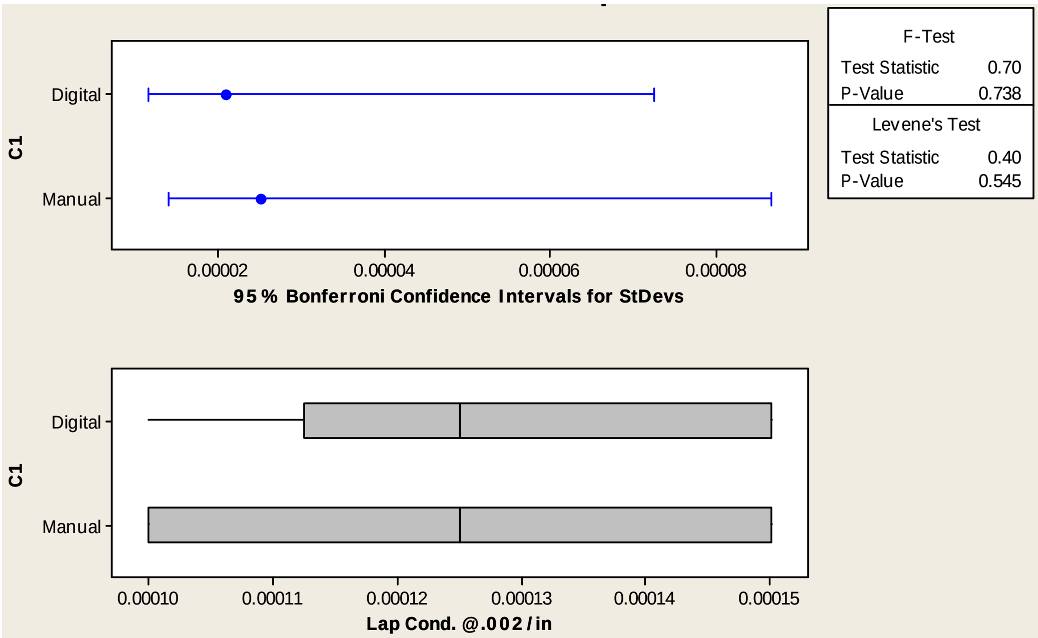


Figure 25. P/N 22253726-104 Test for Equal Variances – Lap Condition at 0.002/in.

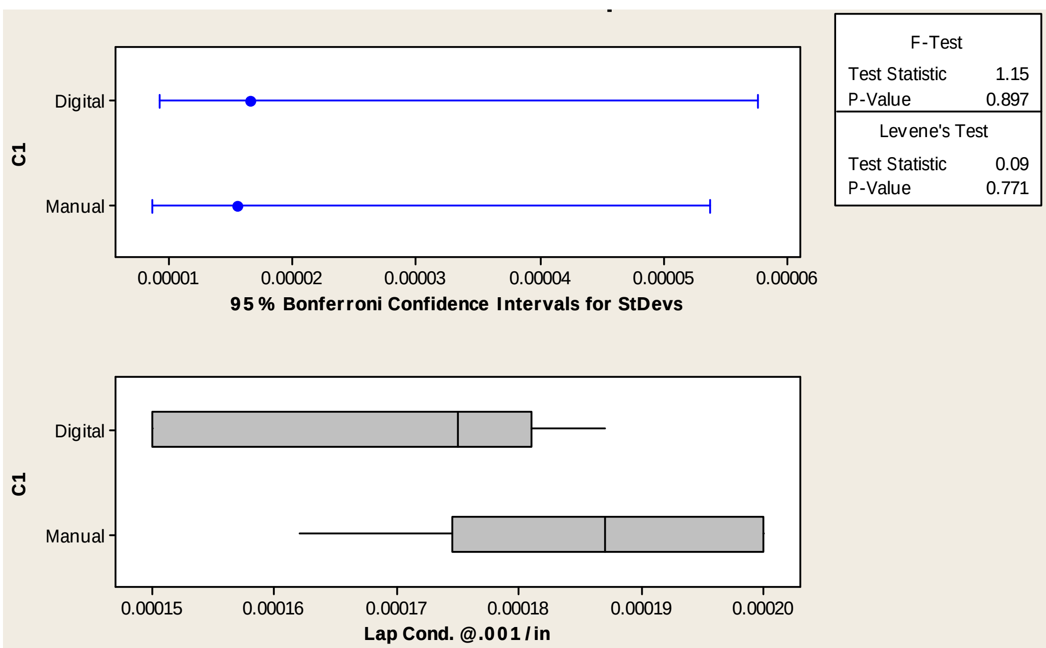


Figure 26. P/N 22253726-104 Test for Equal Variances – Lap Condition at 0.001/in.

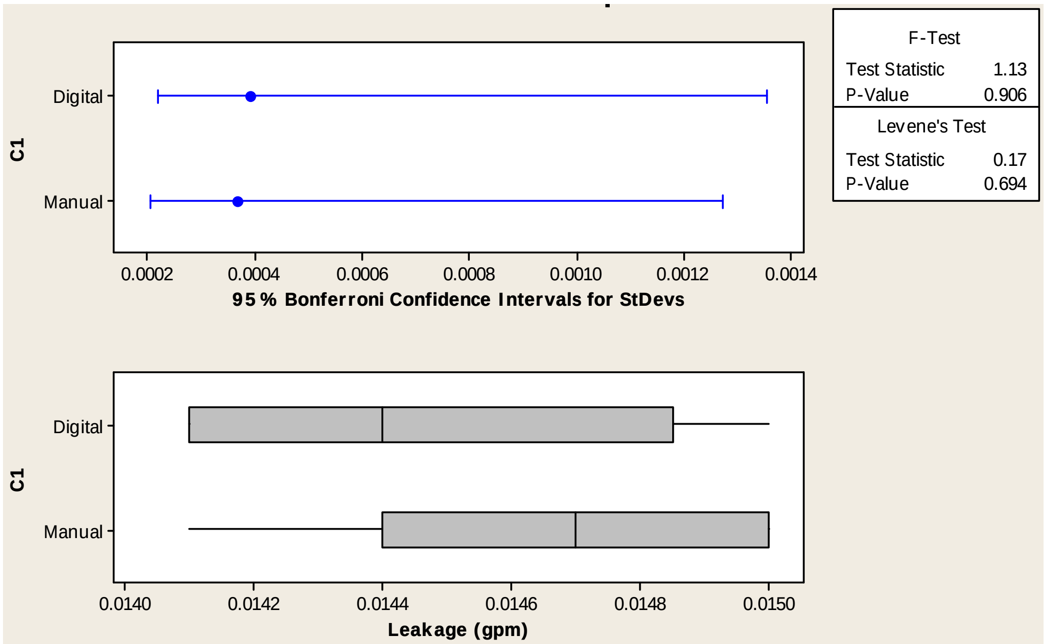


Figure 27. P/N 22253726-104 Test for Equal Variances – Leakage

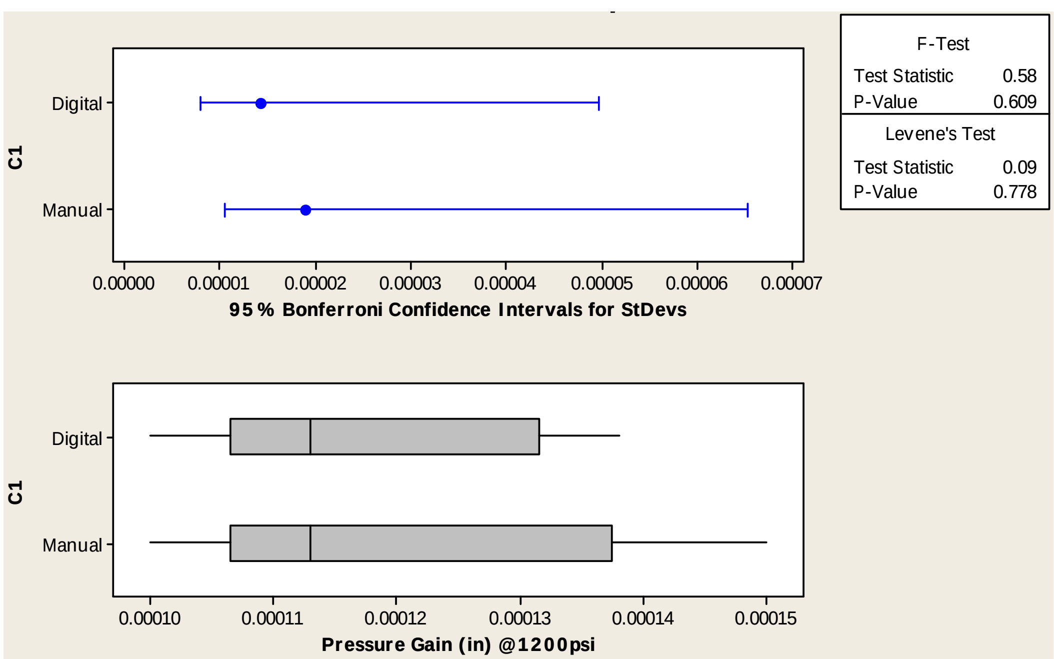


Figure 28. P/N 22253726-104 Test for Equal Variances – Pressure Gain at 1200 psi

Table 1. Flow Grind Data for P/N 28007168-008, 22282790-104, S/N 777

X&Y Recorder					
	R1	R2	R3	R4	R5
Lap Cond. at 0.002/in.	0.000350	0.000325	0.000300	0.000325	0.000300
Lap Cond. at 0.001/in.	0.000375	0.000337	0.000312	0.000337	0.000300
Pressure Gain (in.) at 1200 psi	0.000375	0.000368	0.000368	0.000350	0.000375
Leakage (gpm)	0.2062	0.2025	0.2112	0.2087	0.2075
No-load Drift (psi)	1493	1450	1481	1481	1537
Computer DAQ System					
	D1	D2	D3	D4	D5
Lap Cond. at 0.002/in.	0.000300	0.000300	0.000300	0.000275	0.000300
Lap Cond. at 0.001/in.	0.00287	0.000300	0.000300	0.000300	0.000300
Pressure Gain (in.) at 1200 psi	0.000356	0.000344	0.000350	0.000338	0.000344
Leakage (gpm)	0.2062	0.2087	0.2100	0.2037	0.2037
No-load Drift (psi)	1493	1481	1456	1456	1450

Table 2. Flow Grind Data for P/N 28009498-103, 22284250-104, S/N 365

X&Y Recorder					
	R1	R2	R3	R4	R5
Lap Cond. at 0.002/in.	0.000700	0.00675	0.000675	0.000675	0.000625
Lap Cond. at 0.001/in.	0.000638	0.000663	0.000675	0.000675	0.000613
Pressure Gain (in.) at 1200 psi	0.000287	0.000281	0.000250	0.000268	0.000275
Leakage (gpm)	0.0487	0.0487	0.0487	0.0462	0.0438
No-load Drift (psi)	962	943	1050	1000	943
Computer DAQ System					
	D1	D2	D3	D4	D5
Lap Cond. at 0.002/in.	0.000700	0.000625	0.000700	0.000700	0.000675
Lap Cond. at 0.001/in.	0.000712	0.000687	0.000712	0.000725	0.000650
Pressure Gain (in.) at 1200 psi	0.000263	0.000287	0.000275	0.000263	0.000263
Leakage (gpm)	0.0450	0.0450	0.0438	0.0487	0.0450
No-load Drift (psi)	981	950	1018	994	981

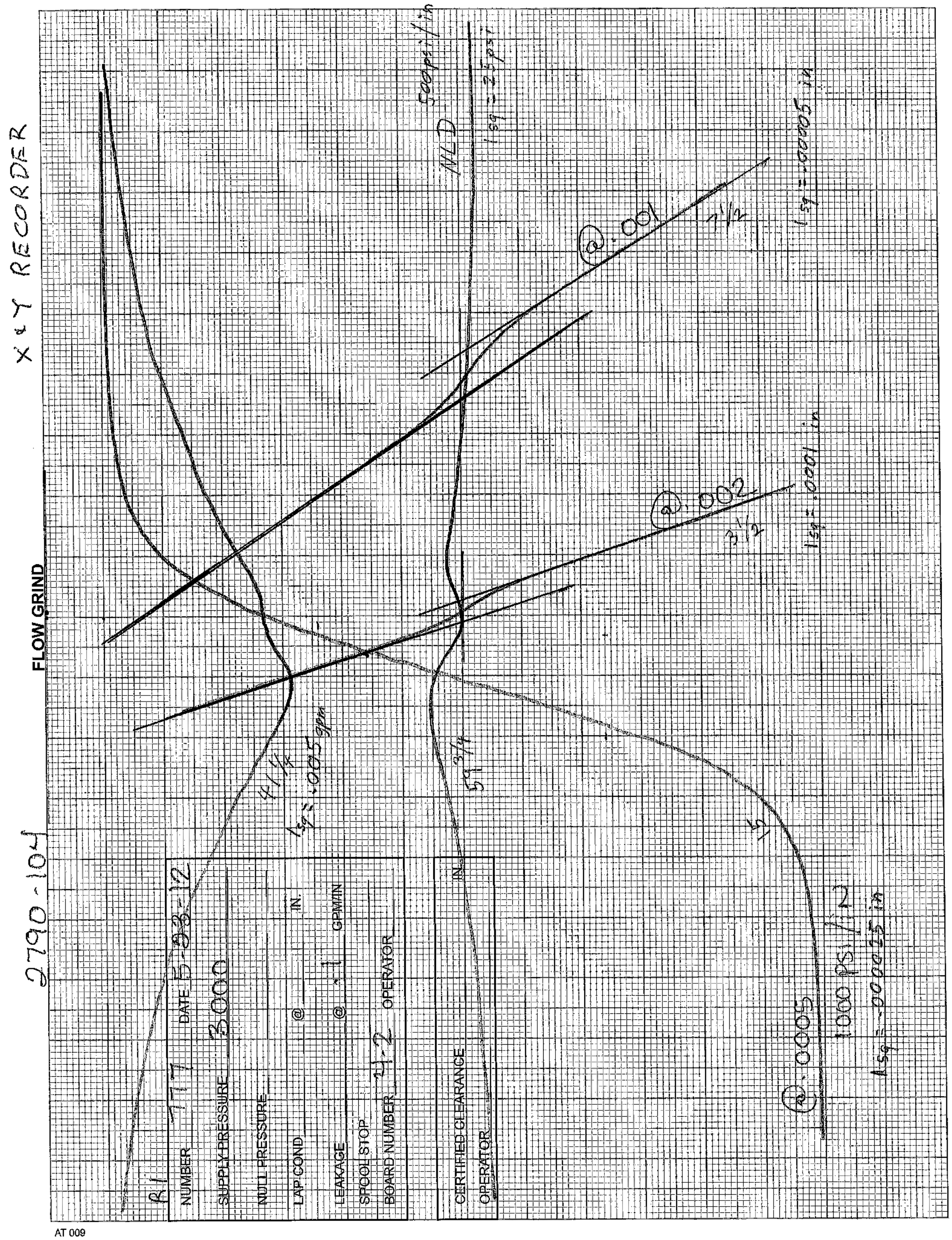
Table 3. Flow Grind Data for P/N 28000747-011, 22253726-104, S/N 835

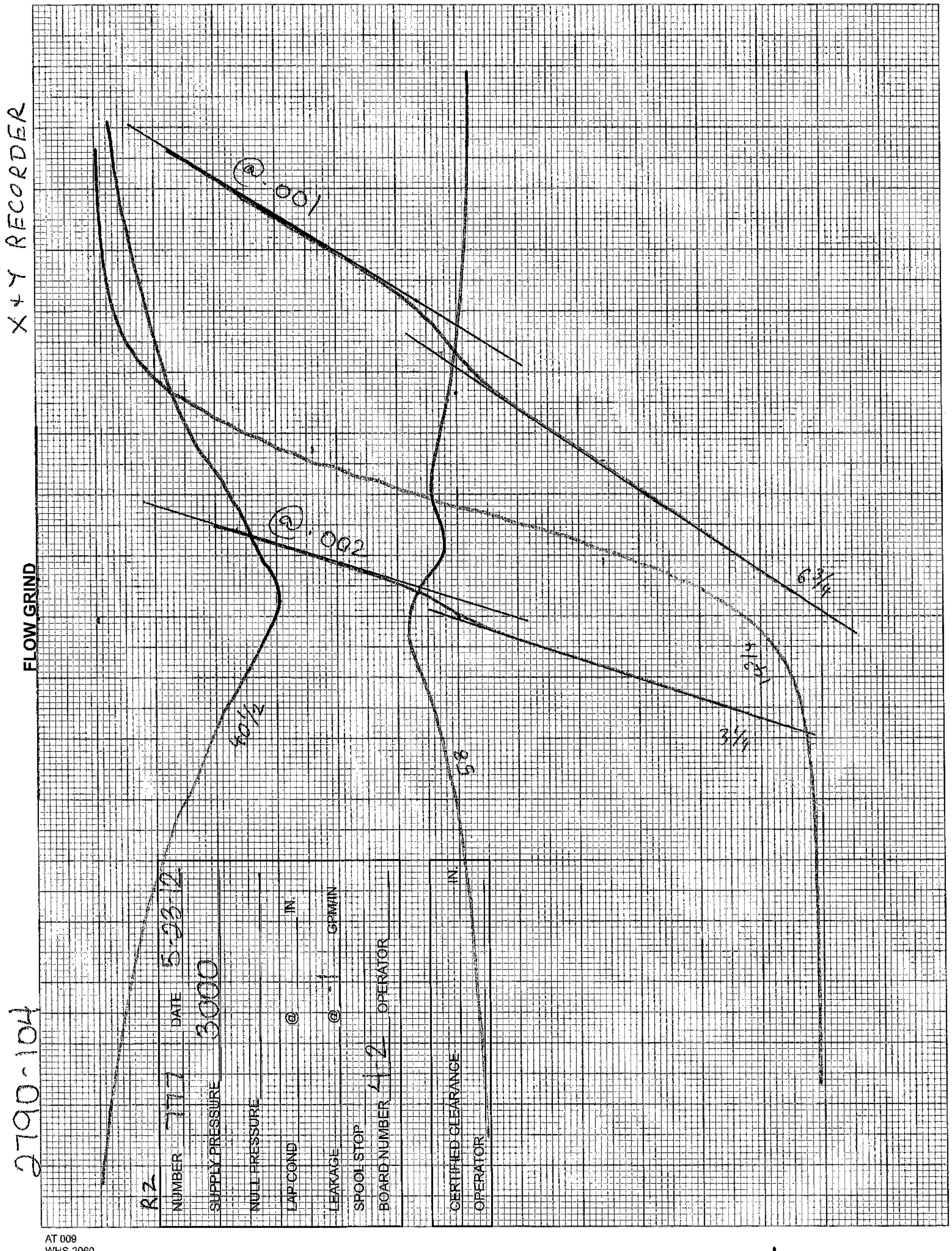
X&Y Recorder					
	R1	R2	R3	R4	R5
Lap Cond. at 0.002/in.	0.000125	0.000150	0.000100	0.000150	0.000100
Lap Cond. at 0.001/in.	0.000187	0.000200	0.000162	0.000200	0.000187
Pressure Gain (in.) at 1200 psi	0.000113	0.000150	0.000125	0.000113	0.000100
Leakage (gpm)	0.0147	0.0150	0.0150	0.0141	0.0147
No-load Drift (psi)	—	—	—	—	—
Computer DAQ System					
	D1	D2	D3	D4	D5
Lap Cond. at 0.002/in.	0.000125	0.000150	0.000150	0.000100	0.000125
Lap Cond. at 0.001/in.	0.000175	0.000175	0.000187	0.000150	0.000150
Pressure Gain (in.) at 1200 psi	0.000113	0.000125	0.000113	0.000100	0.000138
Leakage (gpm)	0.0150	0.0141	0.0141	0.0147	0.0144
No-load Drift (psi)	—	—	—	—	—

APPENDIX A

TEST PLOTS

WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355

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@ .002
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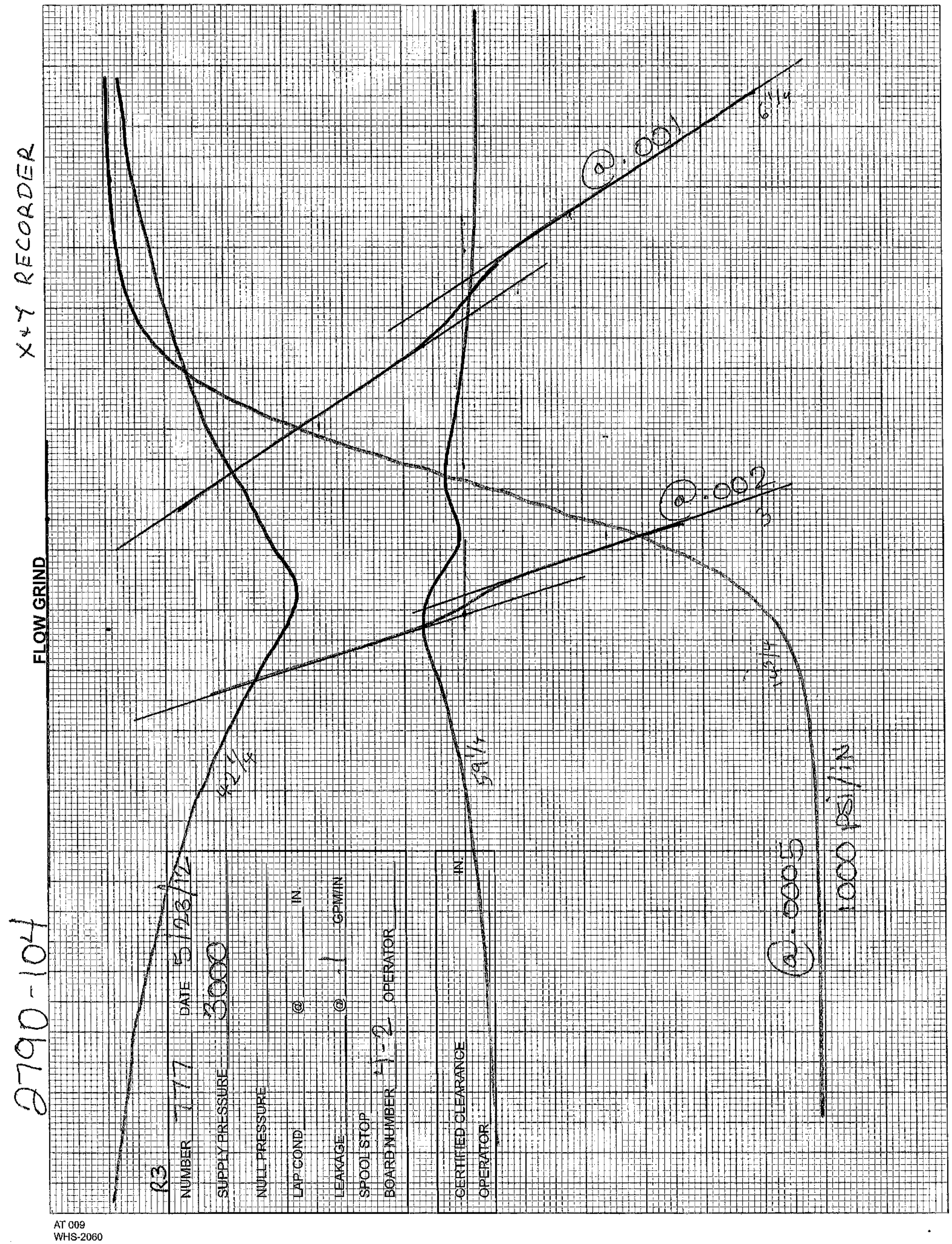
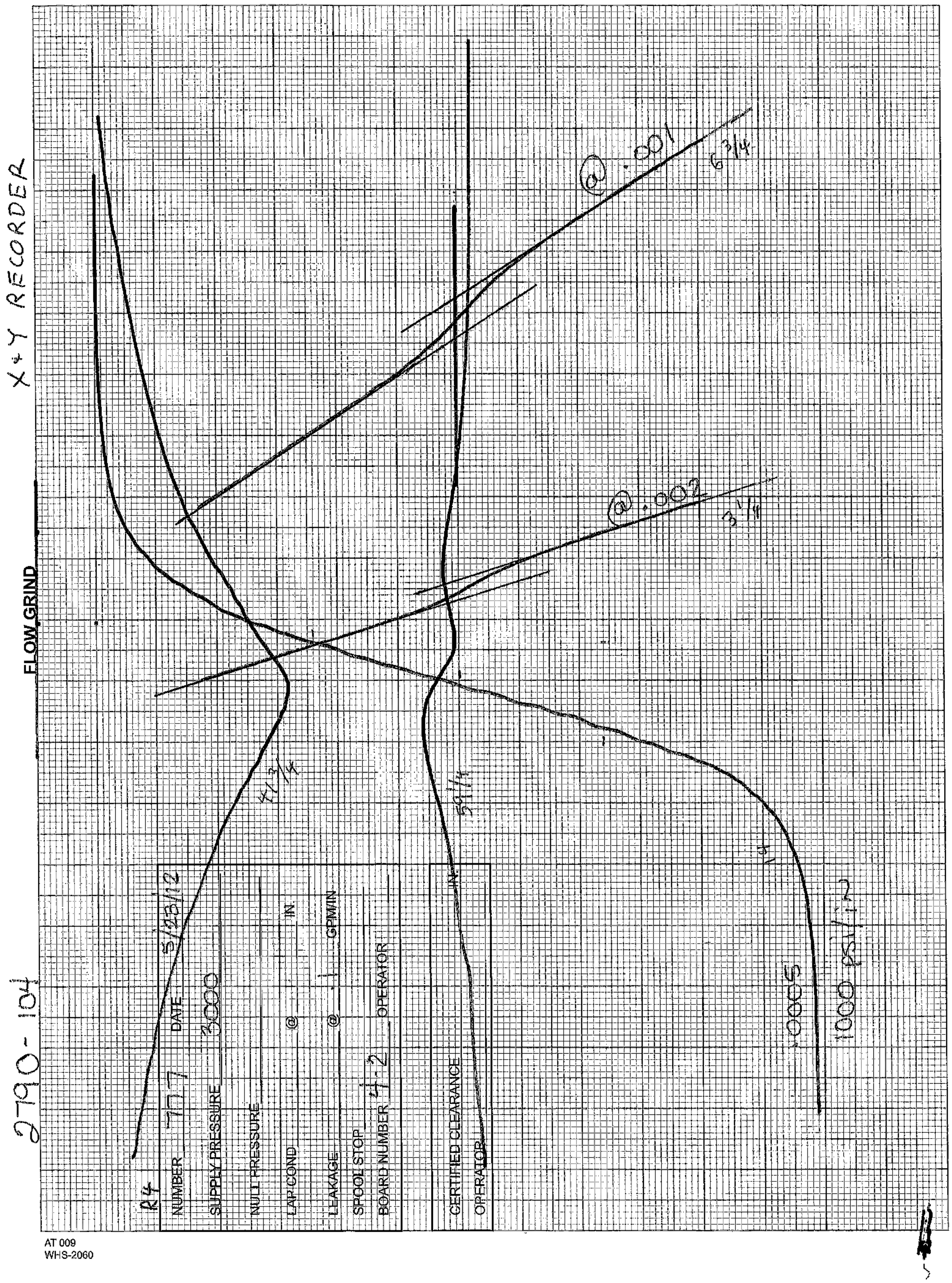
14 3/4
3 1/4
40 1/2
58
R2
NUMBER 777
DATE 5-23-12
SUPPLY PRESSURE 3000
NULL PRESSURE
LAP COND @ IN.
LEAKAGE @ 1 GPM/IN
SPOOL STOP
BOARD NUMBER 4 2
OPERATOR
CERTIFIED CLEARANCE IN.
OPERATOR
AT 009
WHS-2060

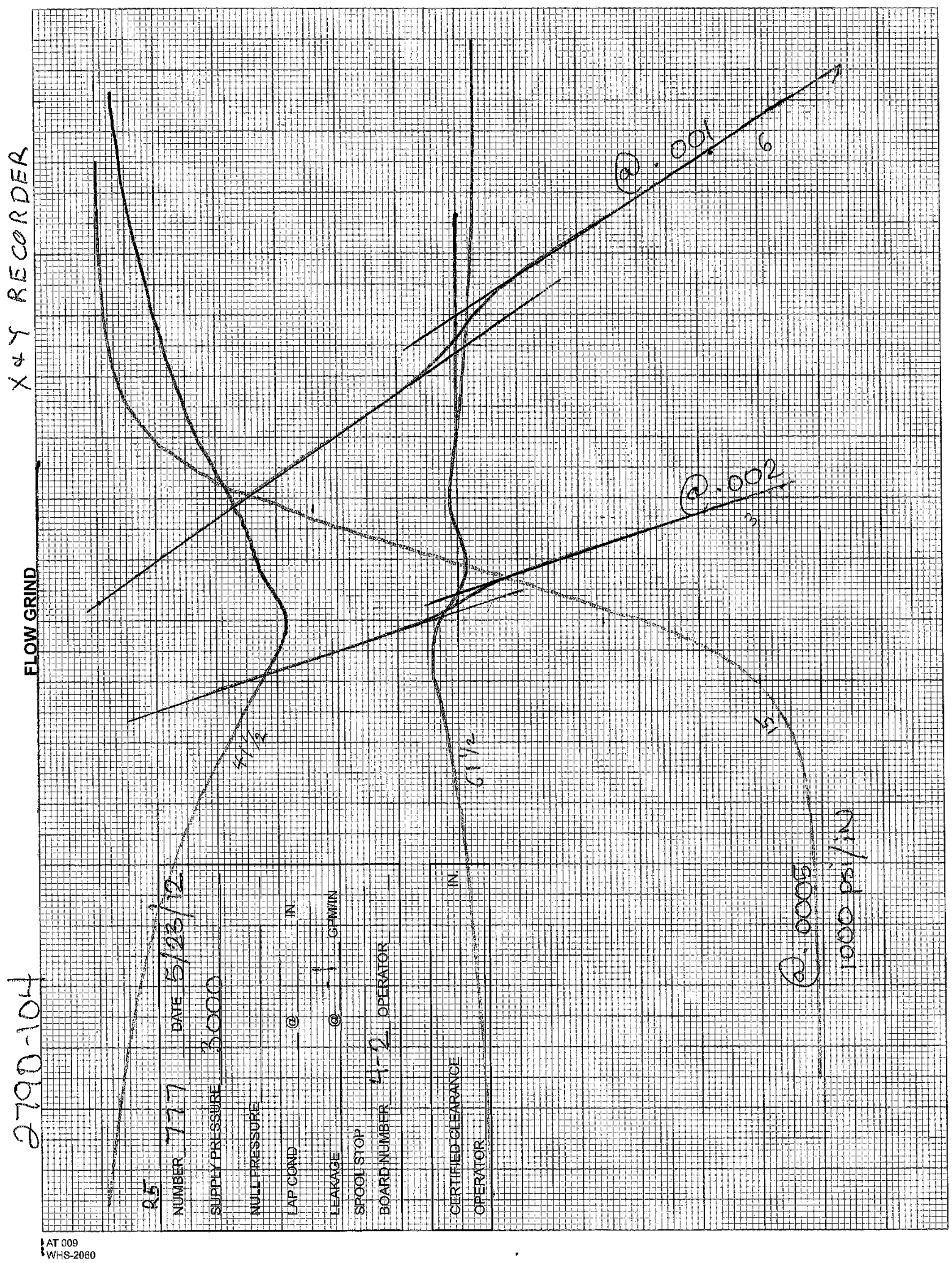
WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355

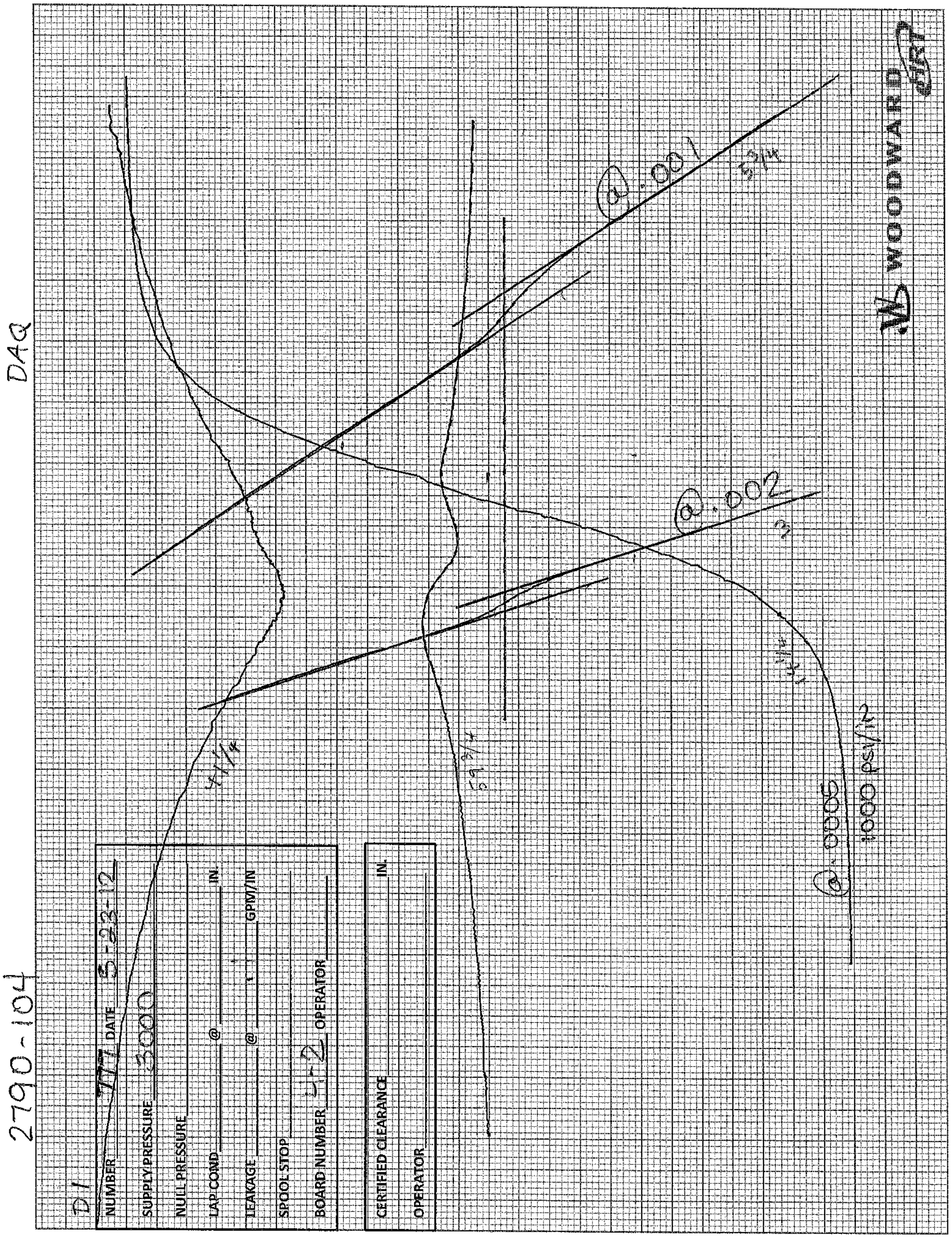


WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355



WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355

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SPOOL STOP
BOARD NUMBER
OPERATOR
CERTIFIED CLEARANCE IN.
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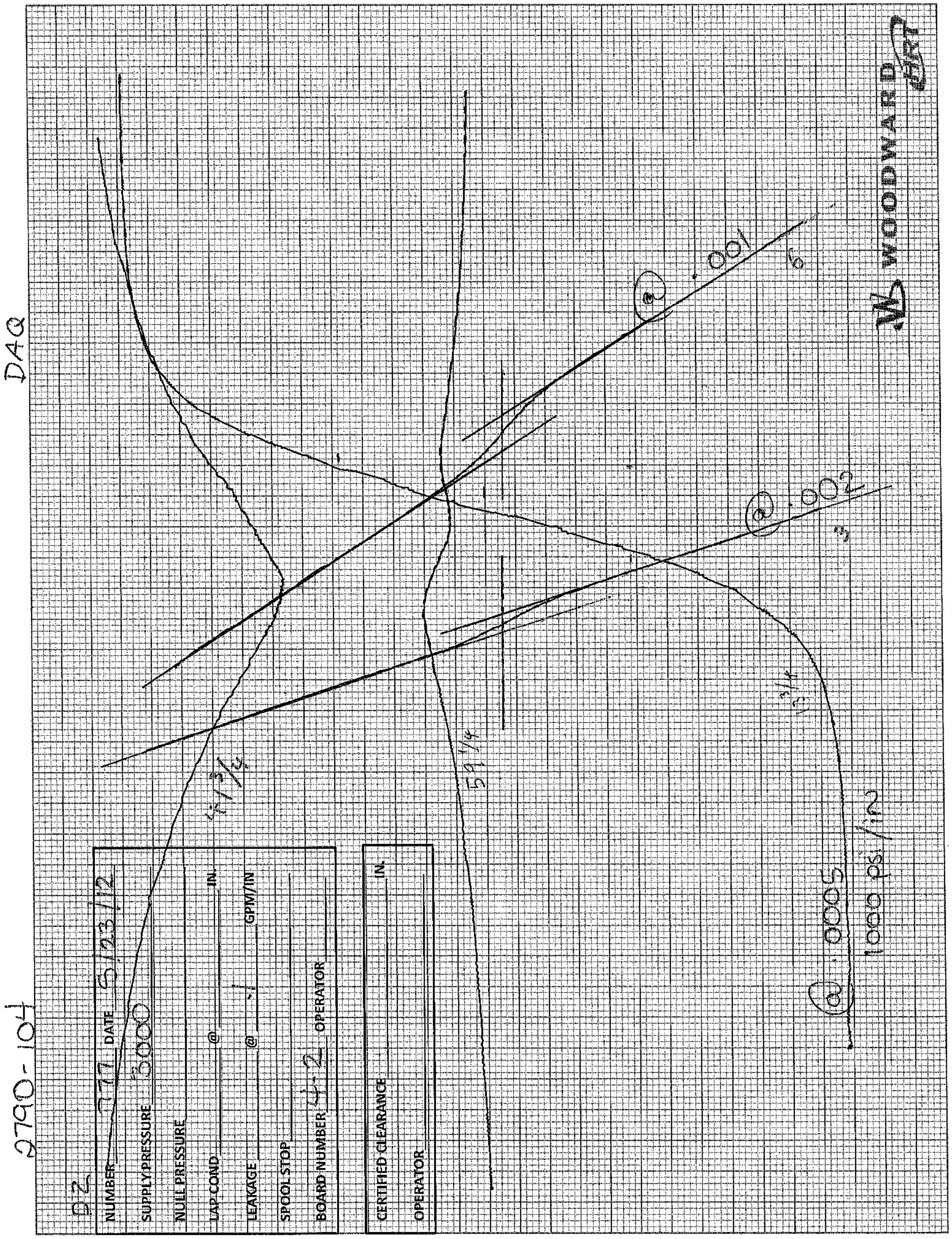
1000 psi/in
WOODWARD HRT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WOODWARD HRT
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NUMBER
DATE 5/23/12
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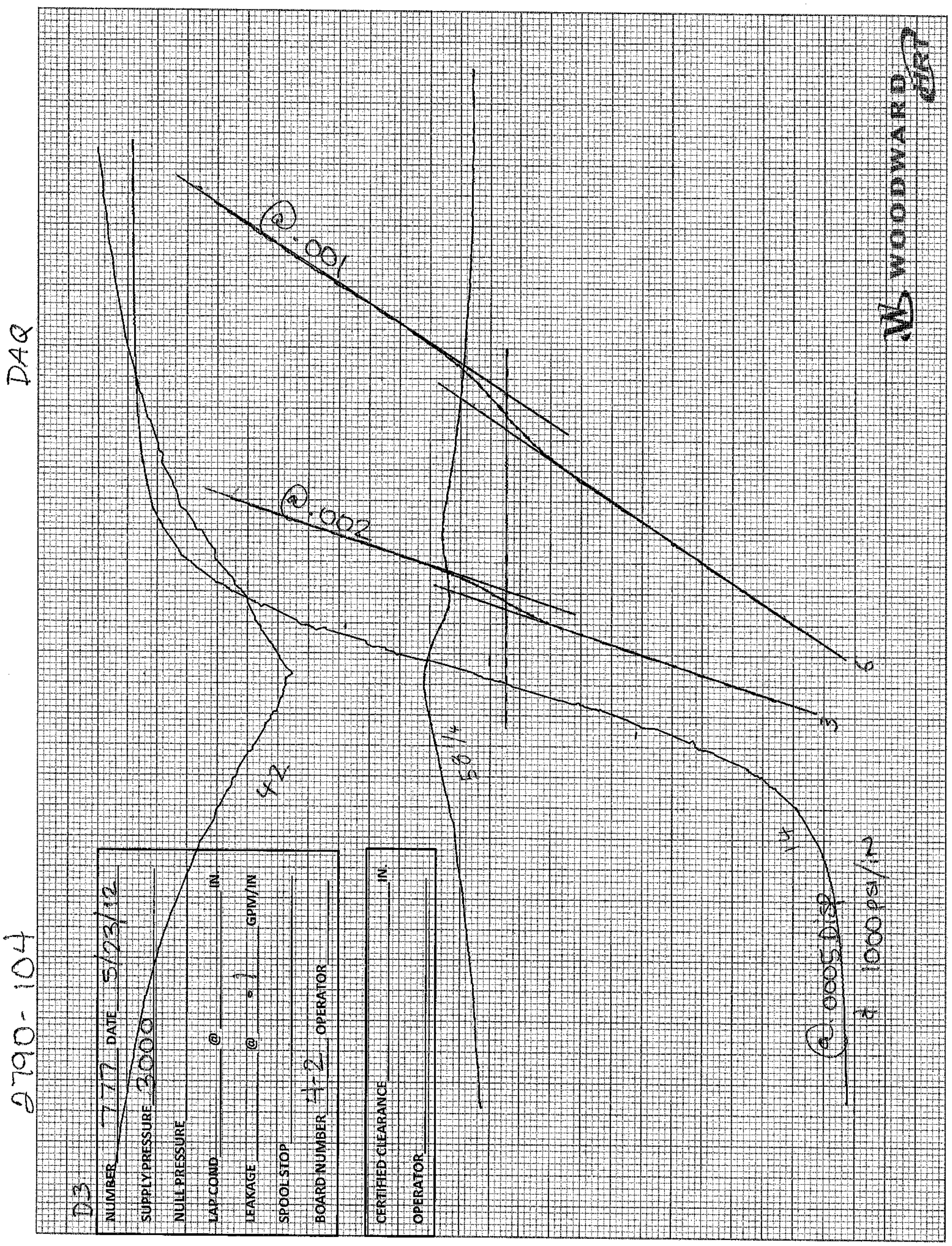
IN.
LEAKAGE
GPM/IN
SPOOL STOP
BOARD NUMBER
OPERATOR
CERTIFIED CLEARANCE
OPERATOR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LEAKAGE @ GPM/IN
SPOOL STOP
BOARD NUMBER 4-2 OPERATOR
CERTIFIED CLEARANCE IN.
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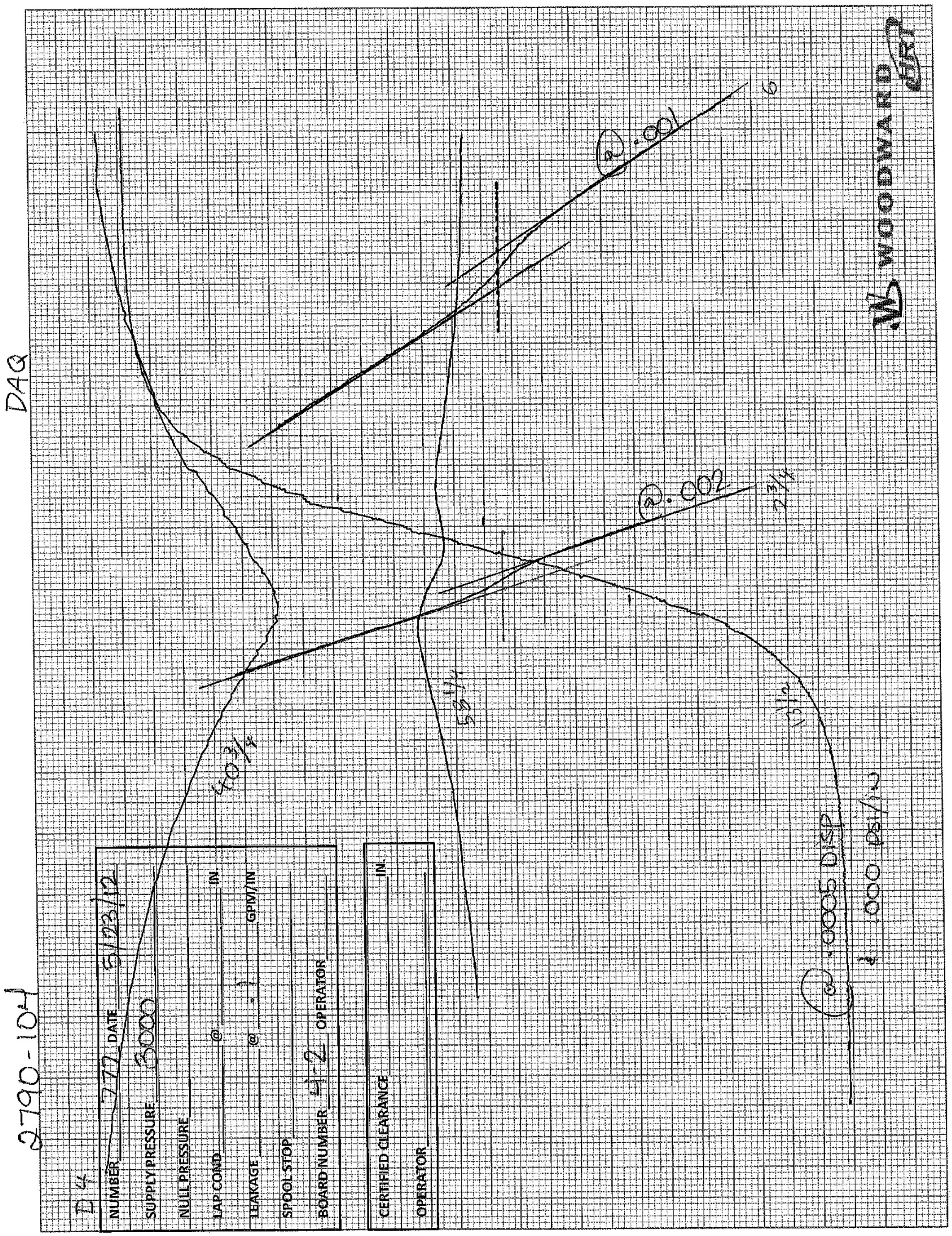
@.001
@.002
@.0005 DISP
1000 PSI/IN
42
WOODWARD HRT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SPOOL STOP
BOARD NUMBER 4-2
OPERATOR
CERTIFIED CLEARANCE IN
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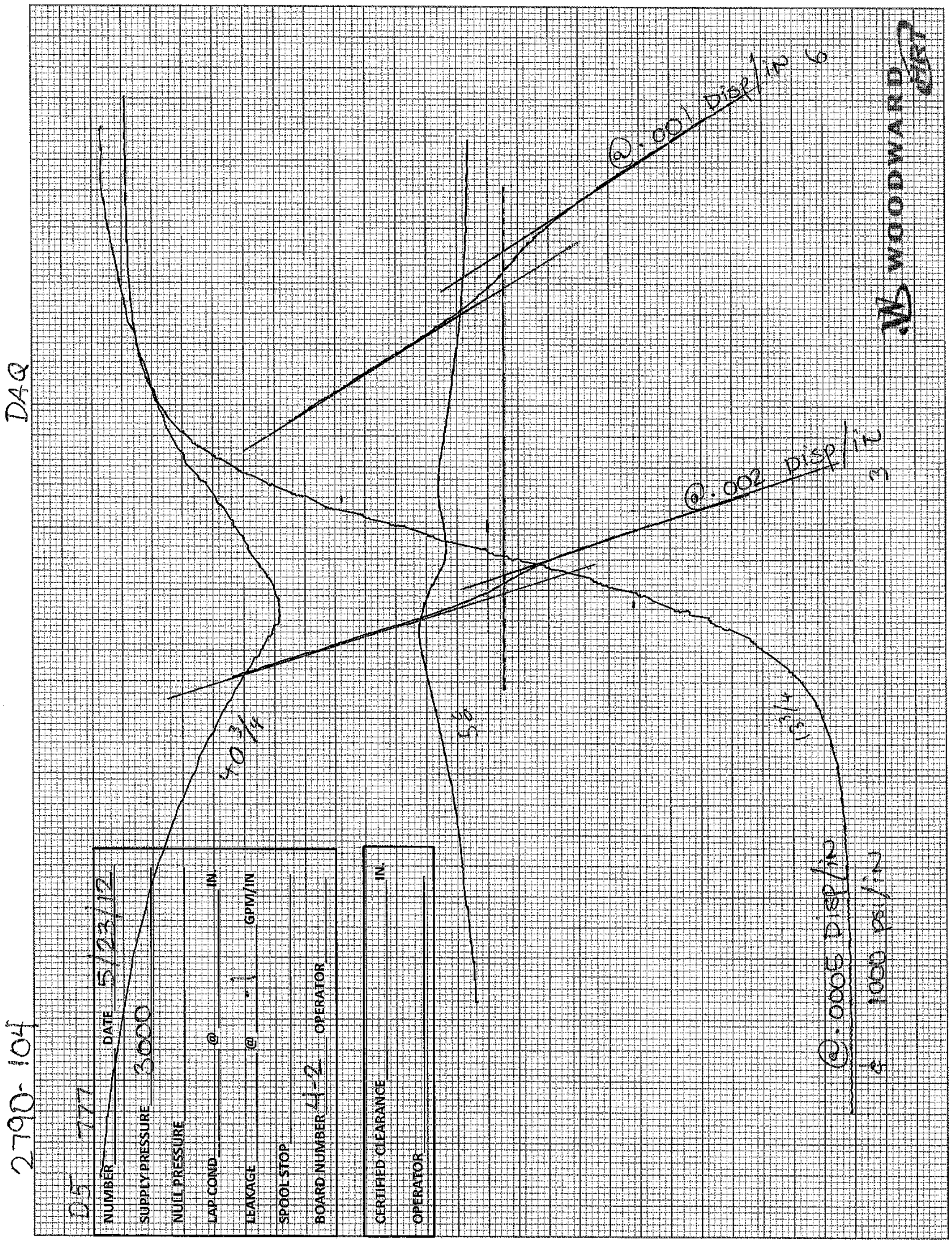
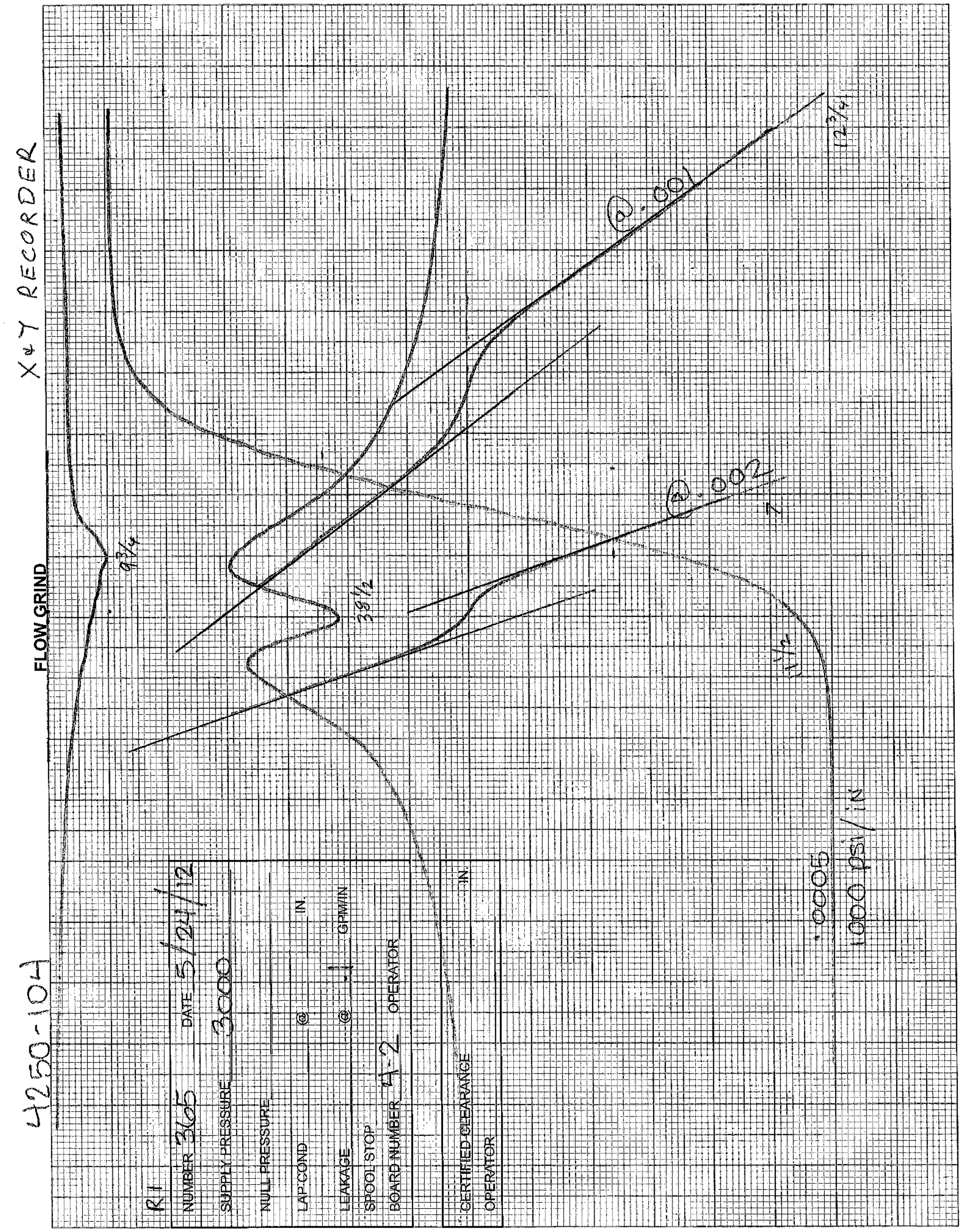
1000 PSI/IN

25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355

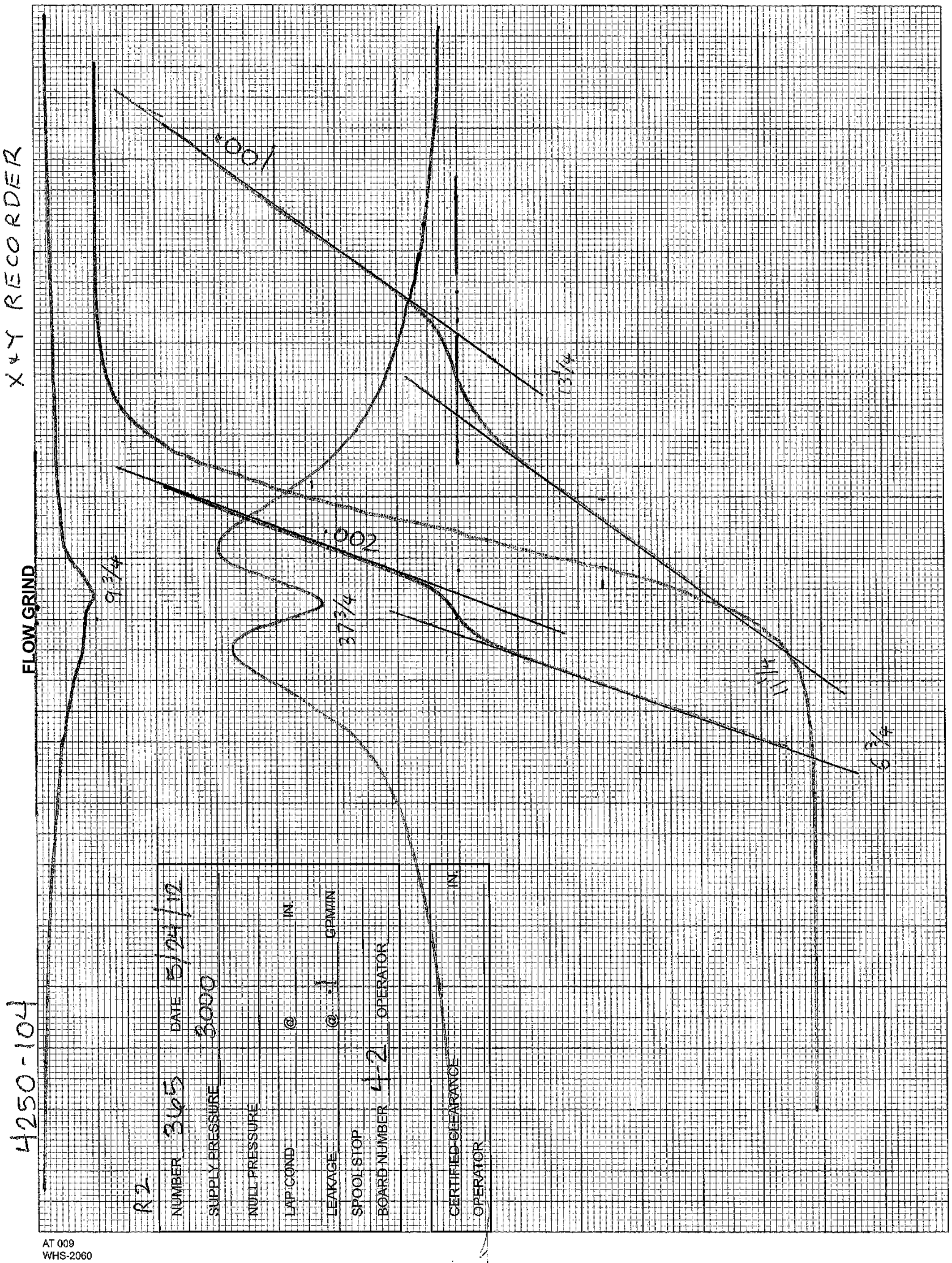


WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355

DOCUMENT NO.: HR70600008



AT 009
WHS-2060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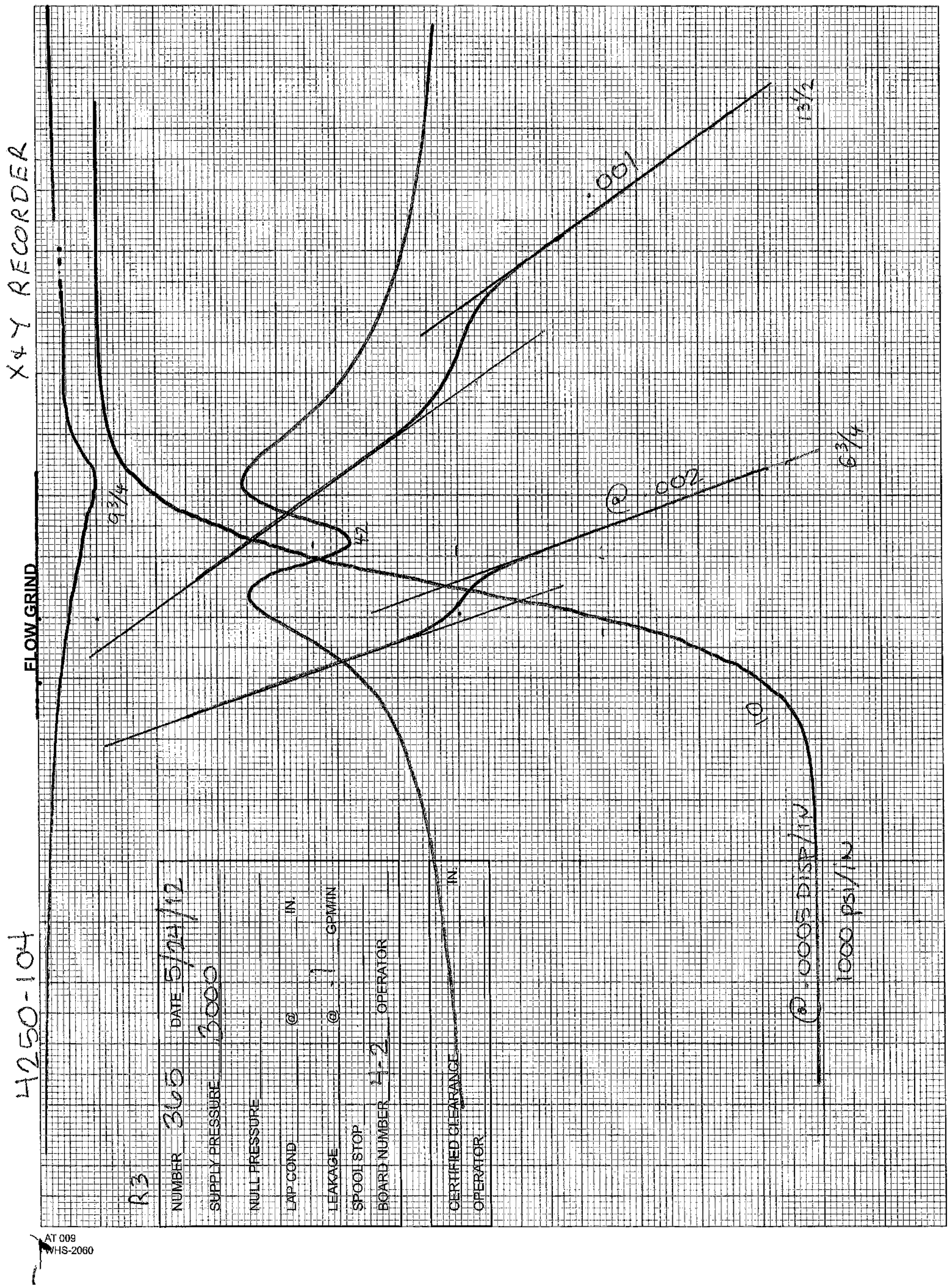
.002
3/4
9 3/4
37 3/4
11 1/4
6 3/4
R2
NUMBER 365
DATE
SUPPLY PRESSURE 3000
NULL PRESSURE
LAP COND @ IN.
LEAKAGE @ .1 GPM/IN
SPOOL STOP
BOARD NUMBER 4-2 OPERATOR
CERTIFIED CLEARANCE IN.
OPERATOR
AT 009
WHS-2060


X+Y RECORDER
4250-104
FLOW GRIND
R3
NUMBER 365 DATE 5/24/12
SUPPLY PRESSURE 3000
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LAP COND @ IN.
LEAKAGE @ .1 GPM/IN
SPOOL STOP
BOARD NUMBER 4-2 OPERATOR
CERTIFIED CLEARANCE IN.
OPERATOR
@ .0005 DISP/IN
1000 PSI/IN
.001
13½
@ .002
6¾
AT 009
WHS-2060

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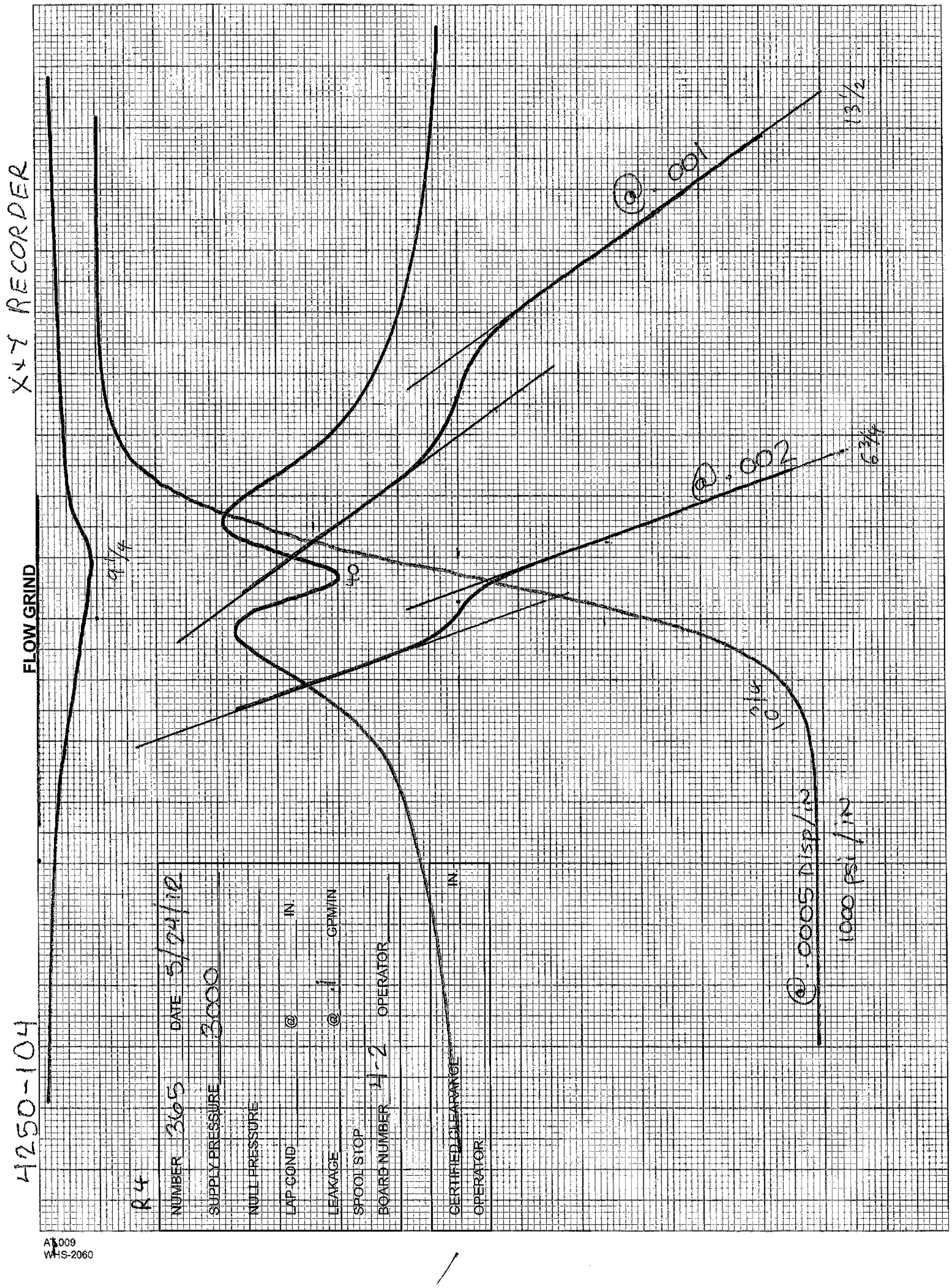
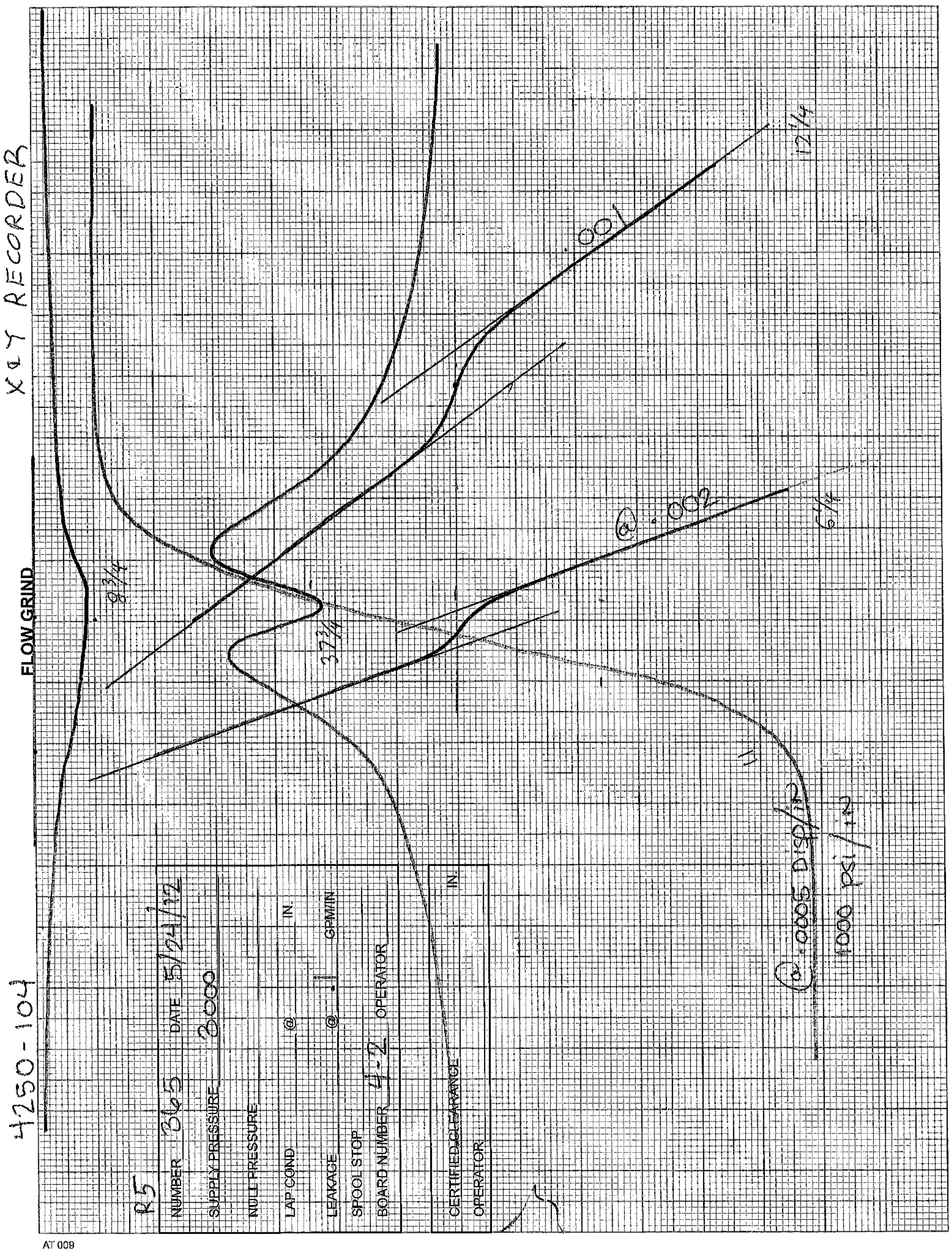


WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355



AT 009
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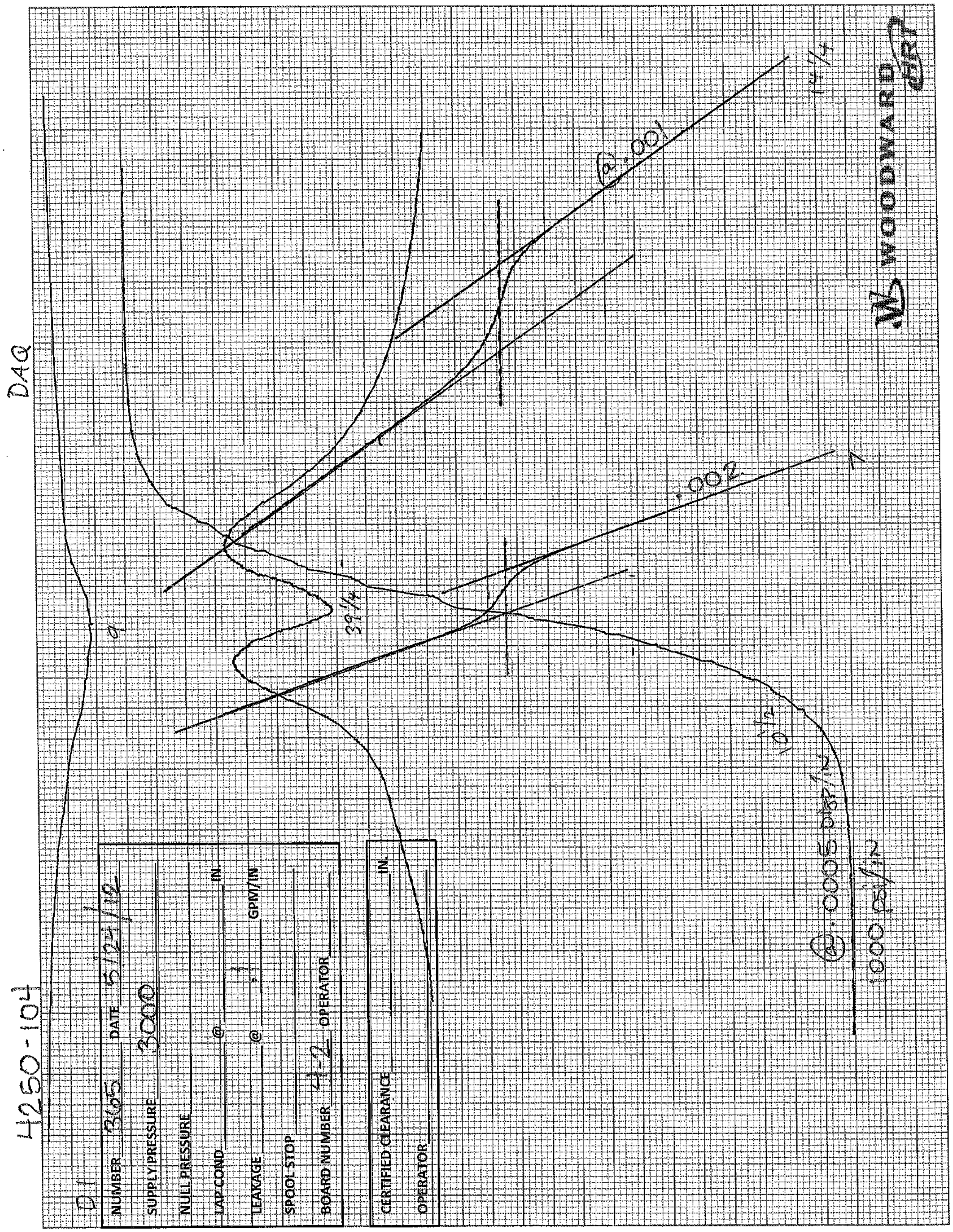
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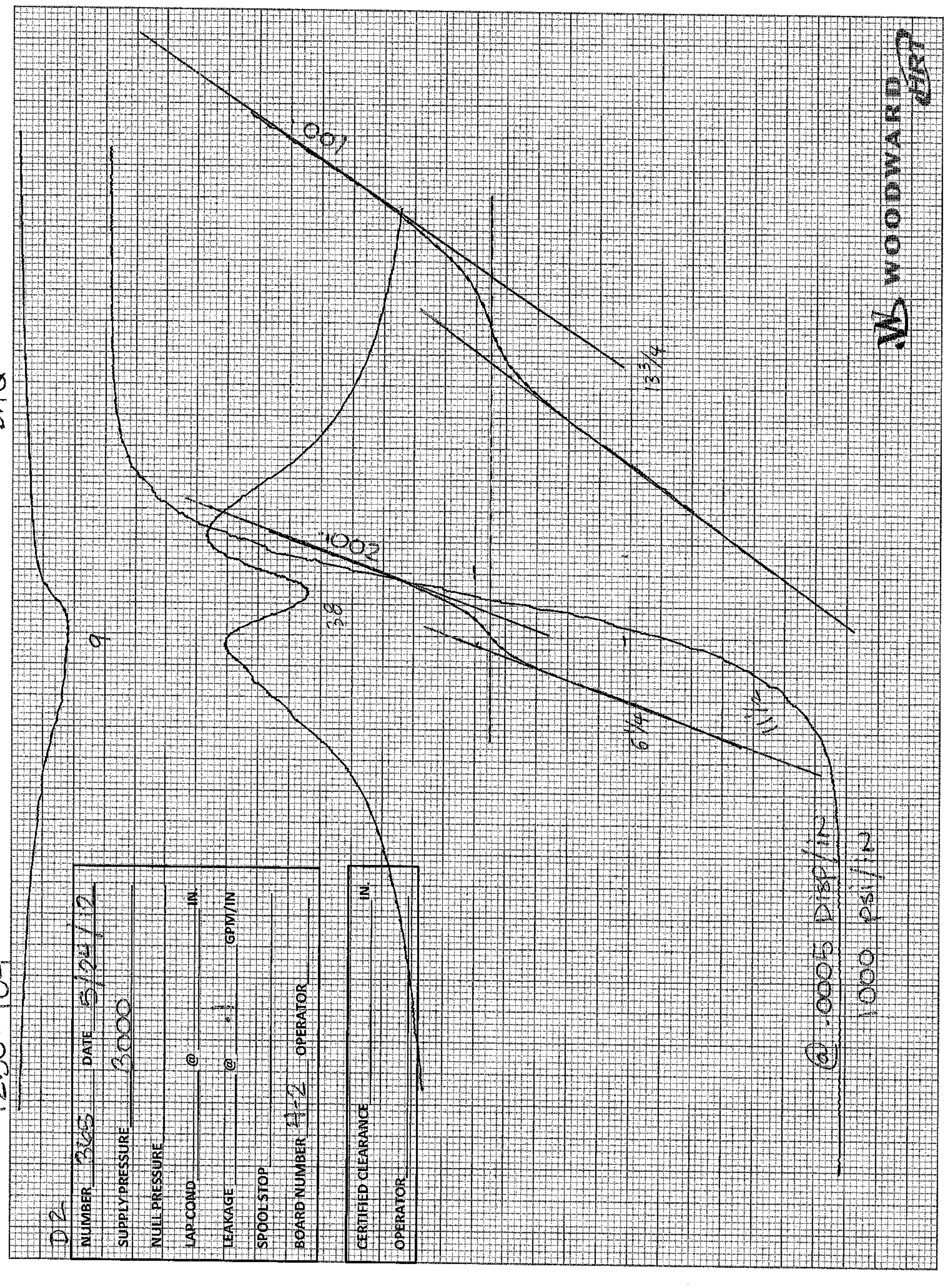
LEAKAGE @ GPM/IN
SPOOL STOP
BOARD NUMBER 4-2
OPERATOR
CERTIFIED CLEARANCE IN
OPERATOR
@ .001
.002
39 1/4
14 1/4
10 1/2
1000 PSI/IN
WOODWARD HRT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DATE 5/24/12
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LEAKAGE @ GPM/IN
SPOOL STOP
BOARD NUMBER A-2
OPERATOR
CERTIFIED CLEARANCE IN.
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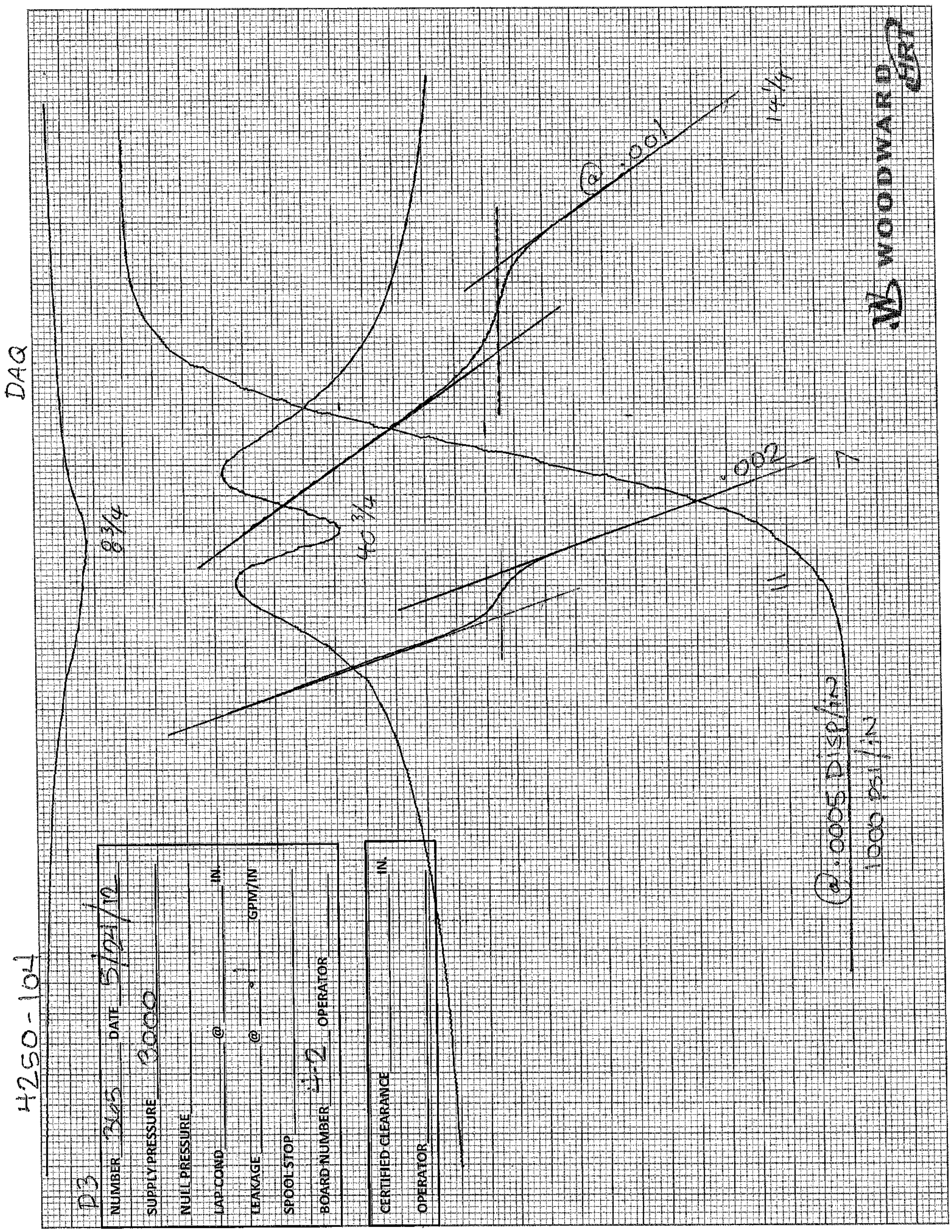
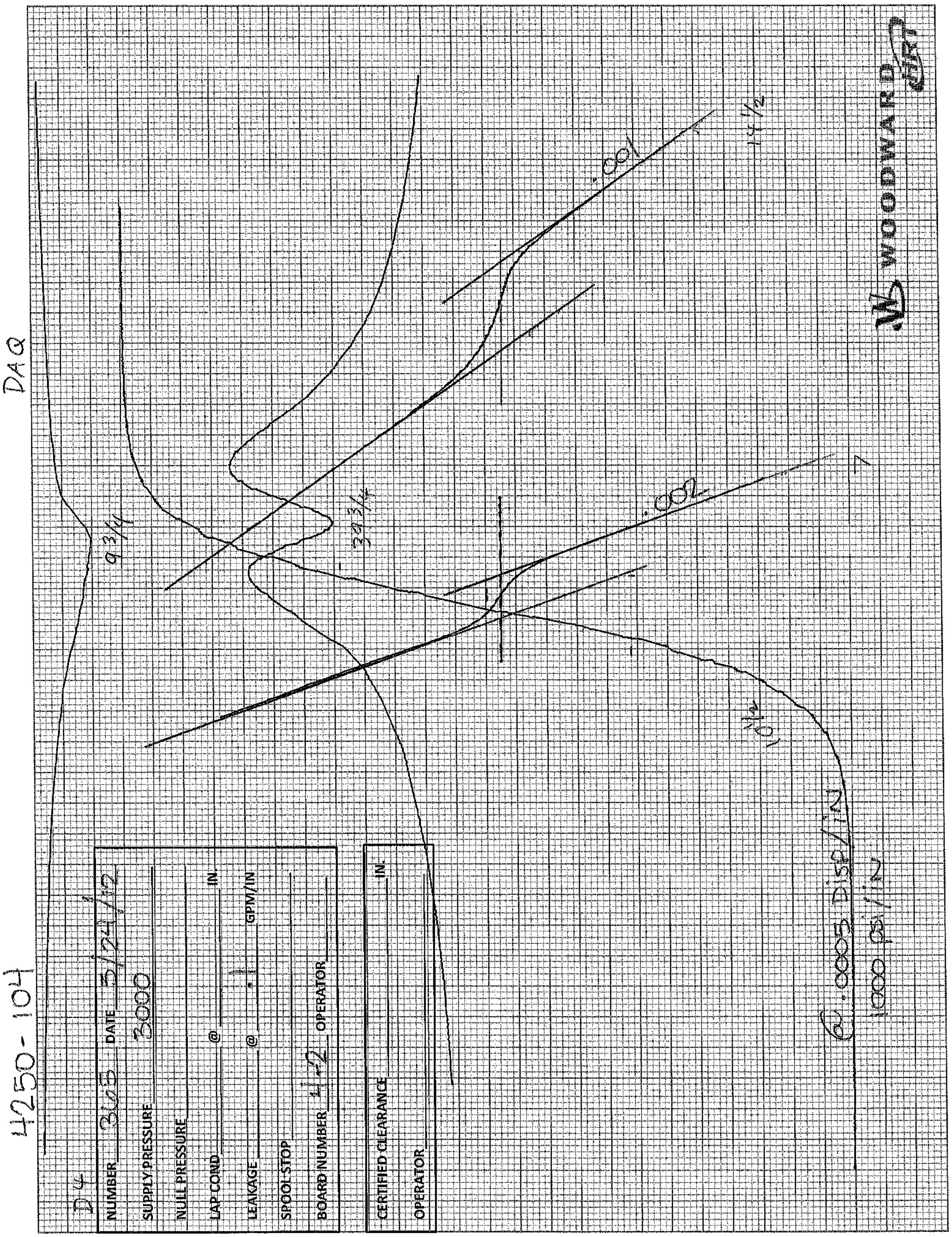
1000 PSI/IN
WOODWARD HRT

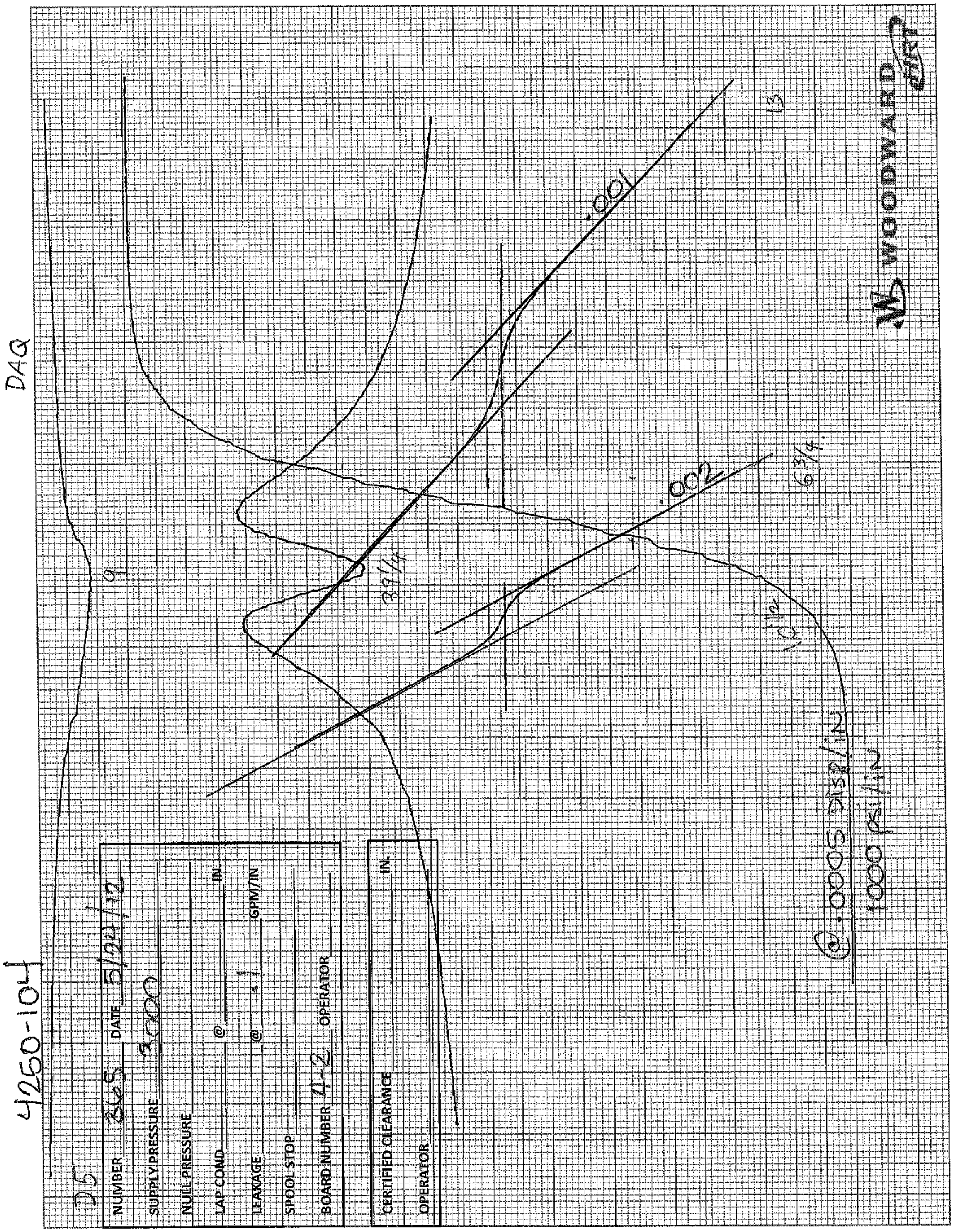
WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355

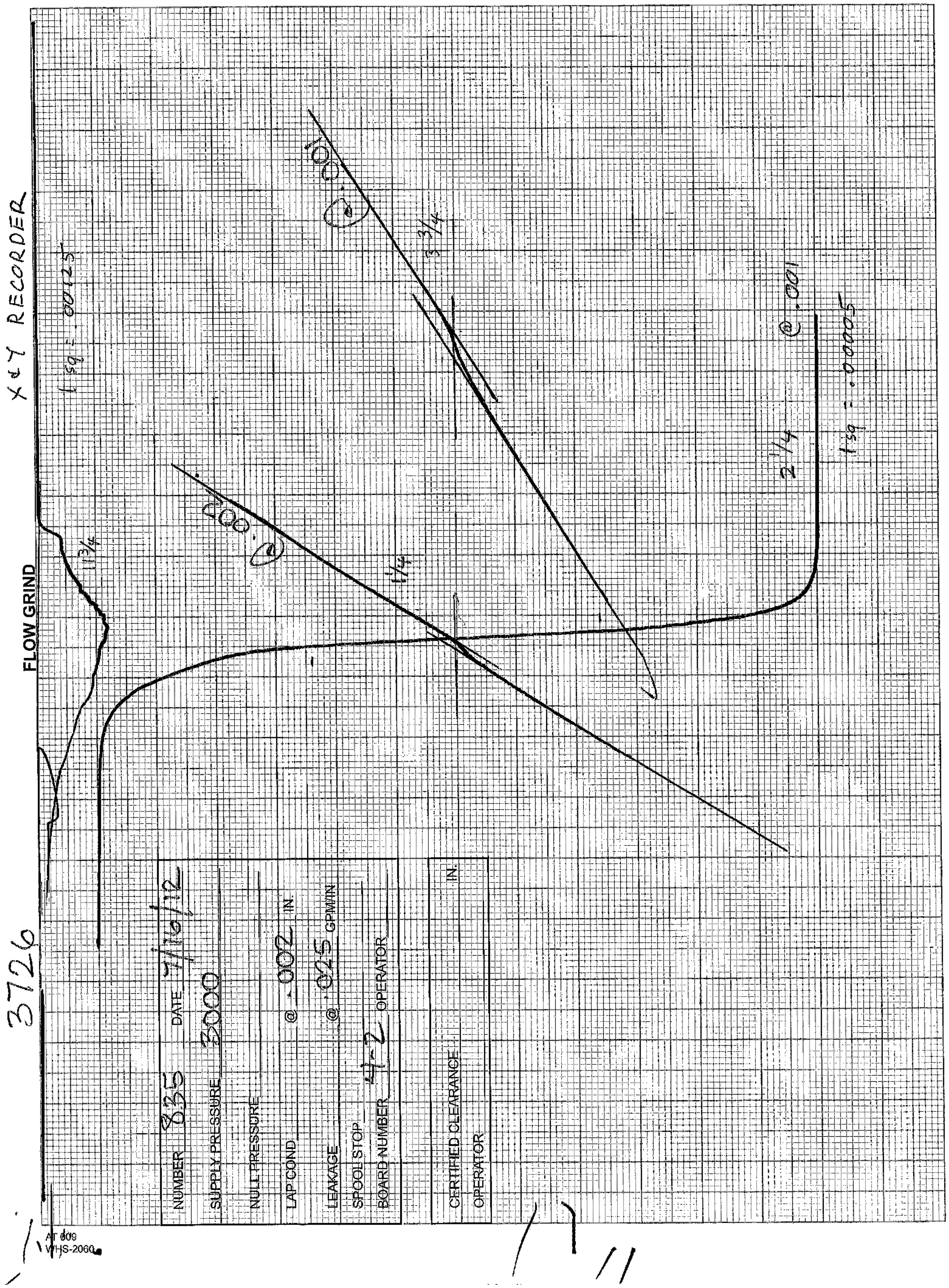


25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355



WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355

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1 3/4
FLOW GRIND
@ .001
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2 1/4
3726
NUMBER 835
DATE 7/10/12
SUPPLY PRESSURE 3000
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LAP COND @ .002 IN.
LEAKAGE @ .025 GPM/IN
SPOOL STOP
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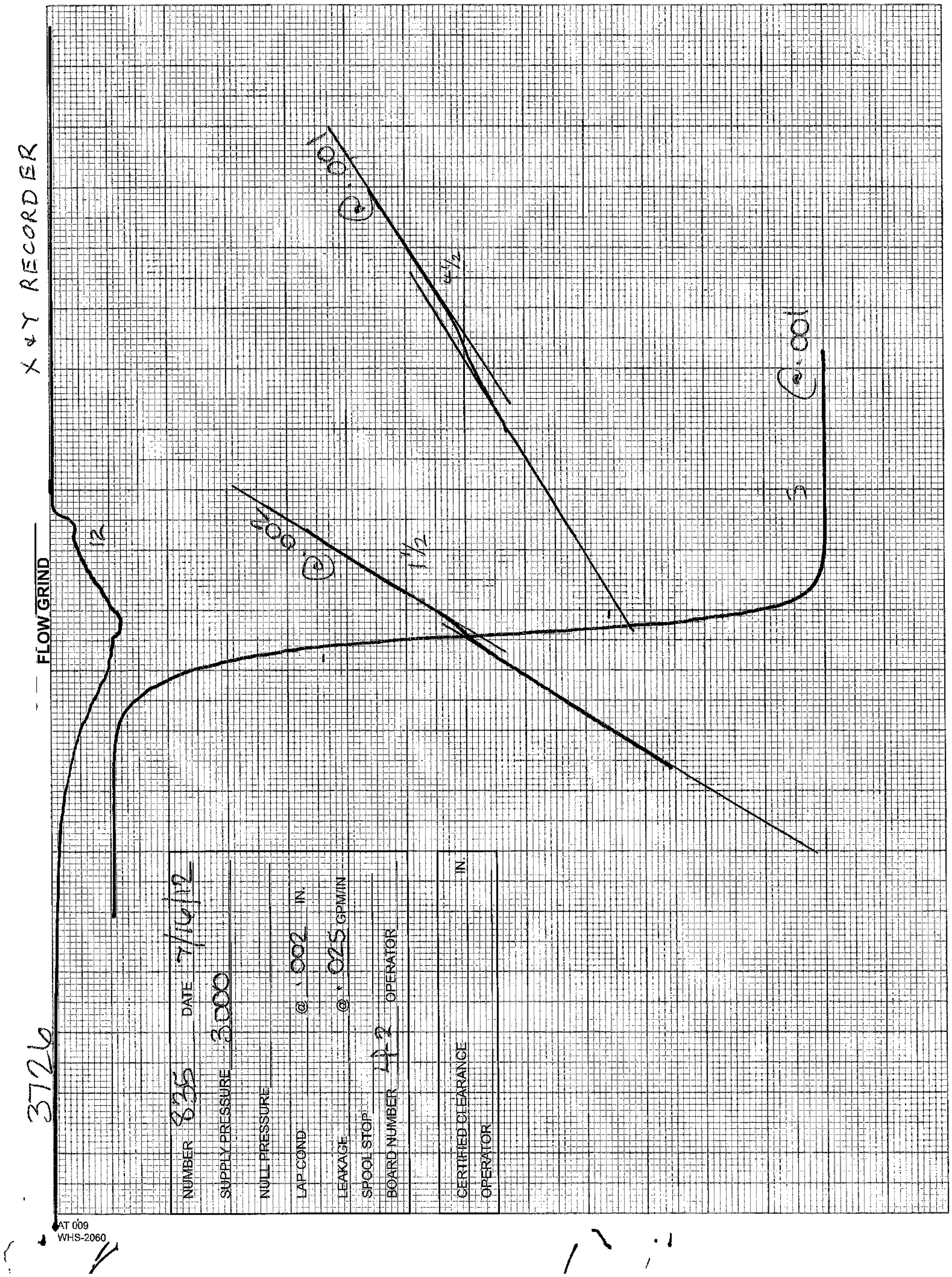
CERTIFIED CLEARANCE IN.
OPERATOR
AT 609
WHS-2060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FLOW GRIND
3726
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SPOOL STOP
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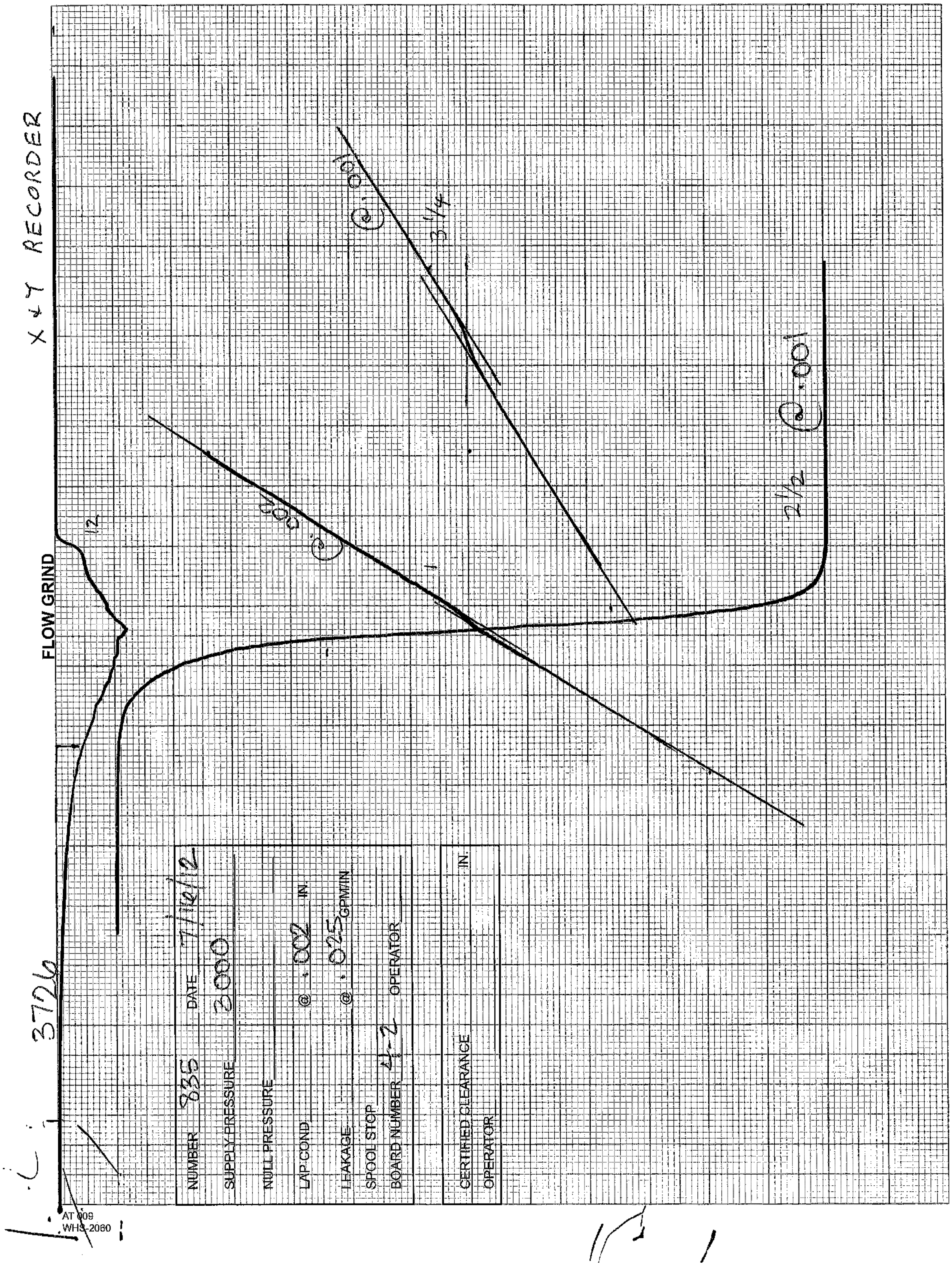
OPERATOR
CERTIFIED CLEARANCE IN.
OPERATOR
AT 009
WHS-2060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FLOW GRIND
3726
@ .001
3 1/4
2 1/2 @ .001
@ .001
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DATE 7/16/12
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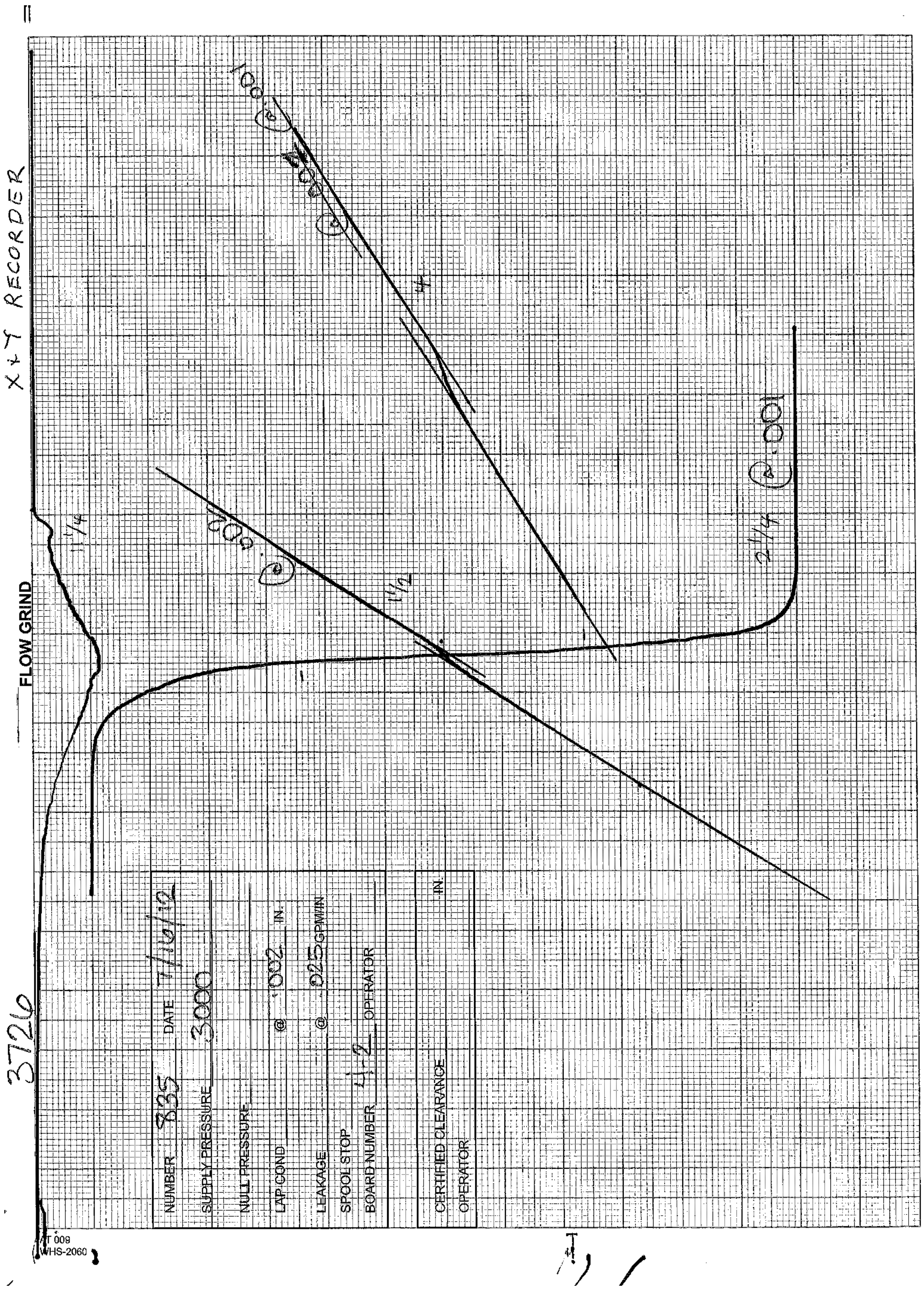
NULL PRESSURE
LAP COND @ .002 IN.
LEAKAGE @ .025 GPM/IN
SPOOL STOP
OPERATOR
BOARD NUMBER 4-2
CERTIFIED CLEARANCE IN.
OPERATOR
AT 009
WHS-2080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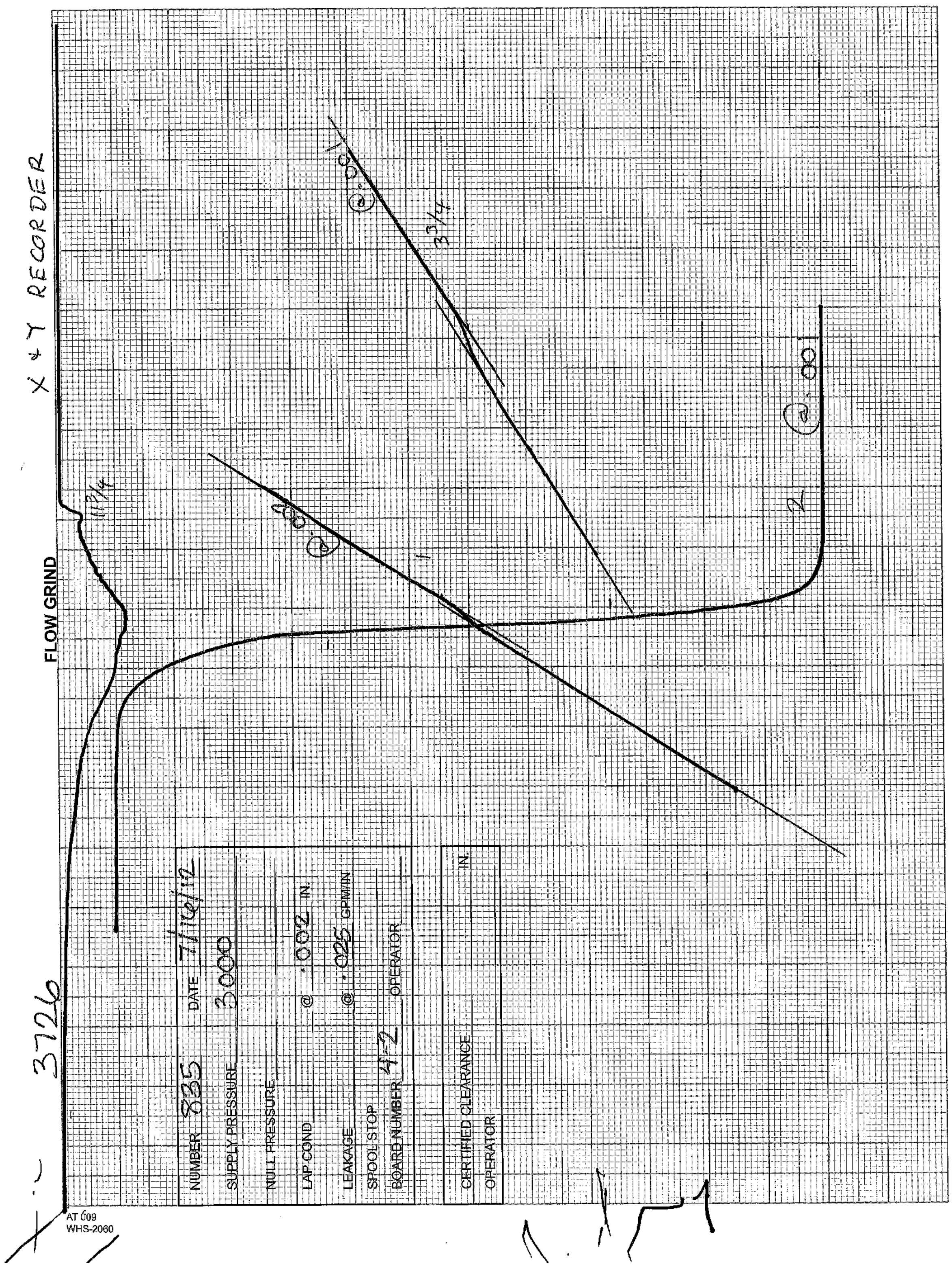
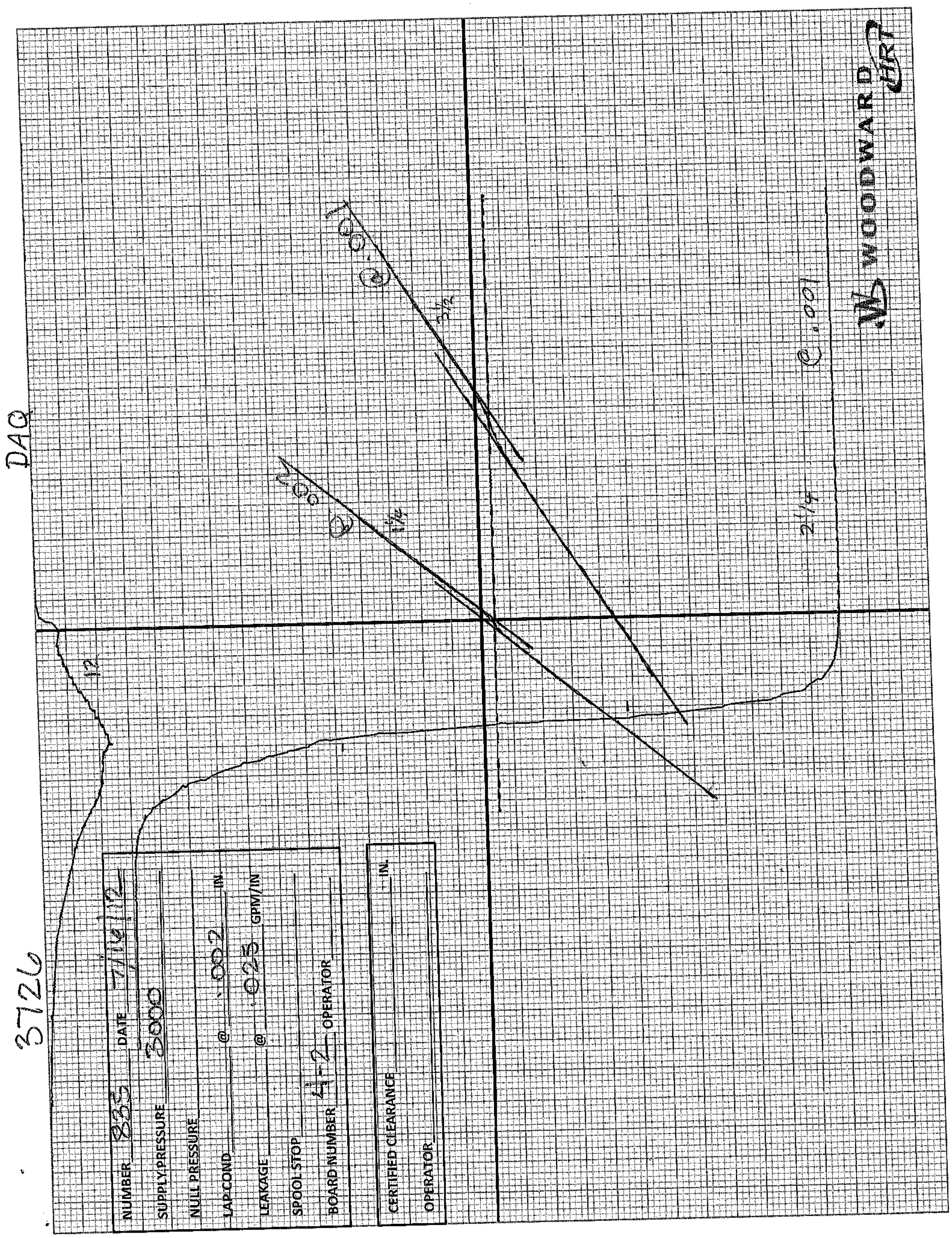
LAP COND @ .002 IN.
LEAKAGE @ .025 GPM/IN
SPOOL STOP
BOARD NUMBER 4-2
OPERATOR
CERTIFIED CLEARANCE IN.
OPERATOR
1 1/4
1 1/2
2 1/4 @ .001
WHS-2060

WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355

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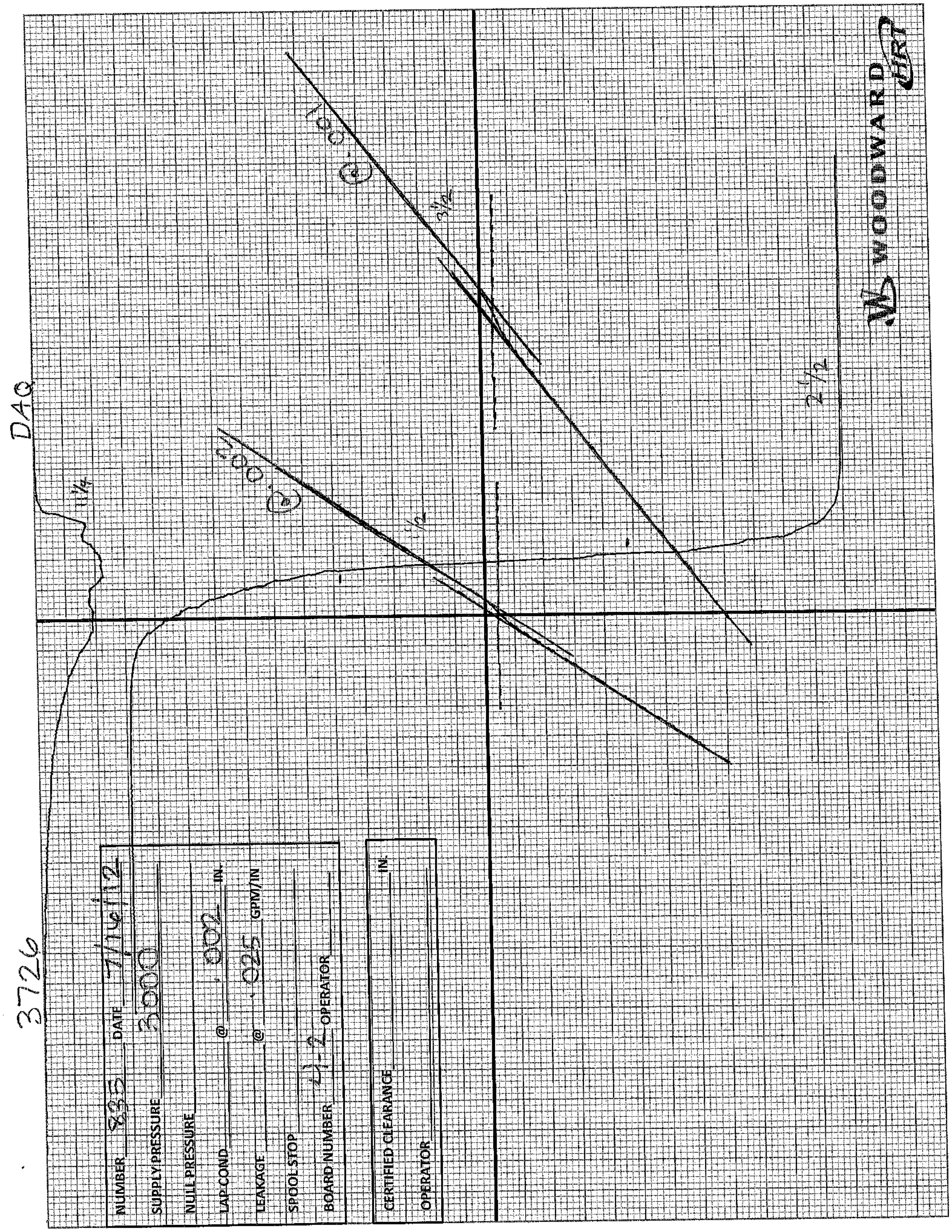
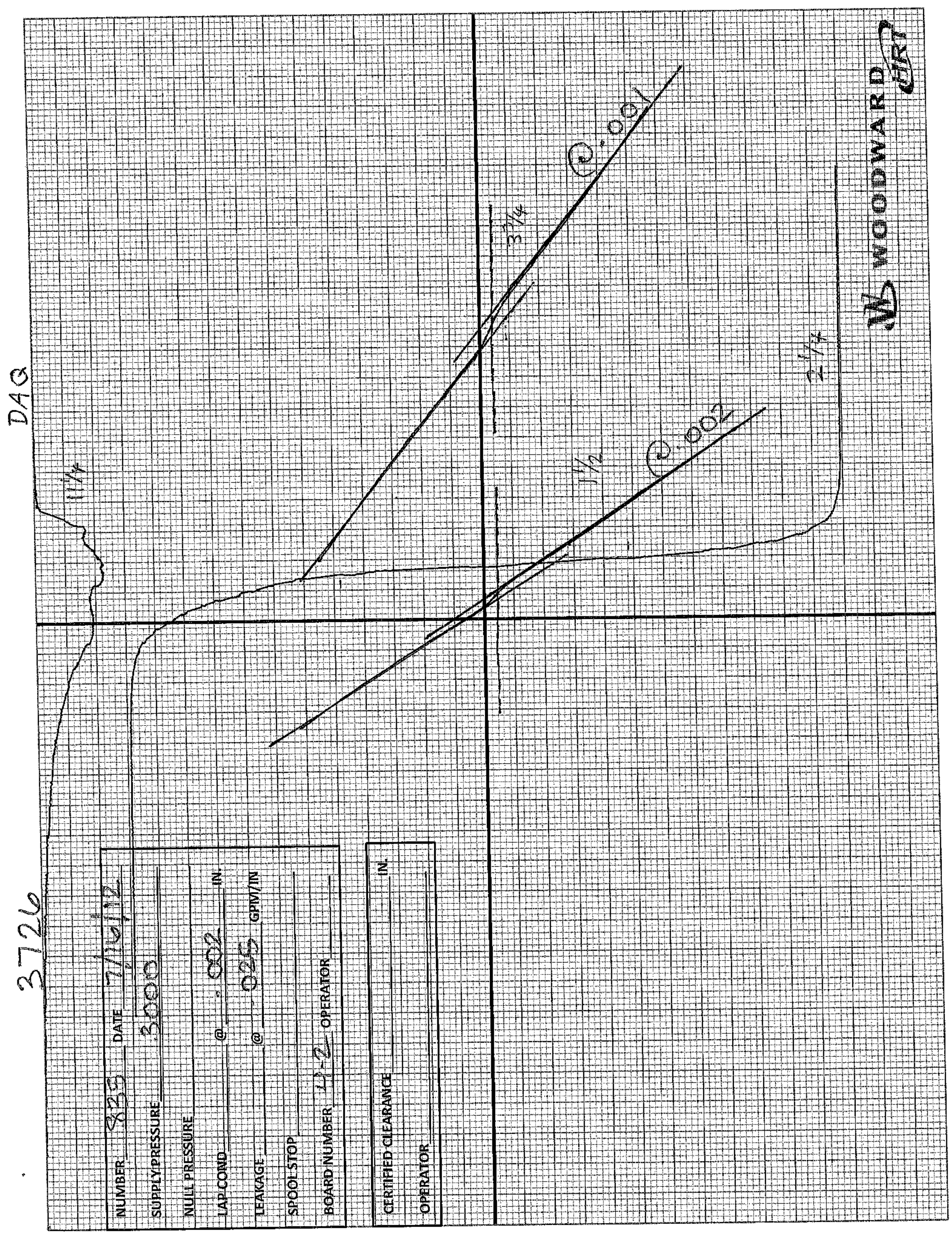
LEAKAGE @ .025 GPM/IN
SPOOL STOP
BOARD NUMBER 4-2
OPERATOR
CERTIFIED CLEARANCE IN.
OPERATOR
WOODWARD HRT
@ .001
2 1/4

WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355

DOCUMENT NO.: HR70600008





WOODWARD HRT
DAG
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3 1/4
@.002
1 1/2
2 1/4
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SPOOL STOP
BOARD NUMBER 42
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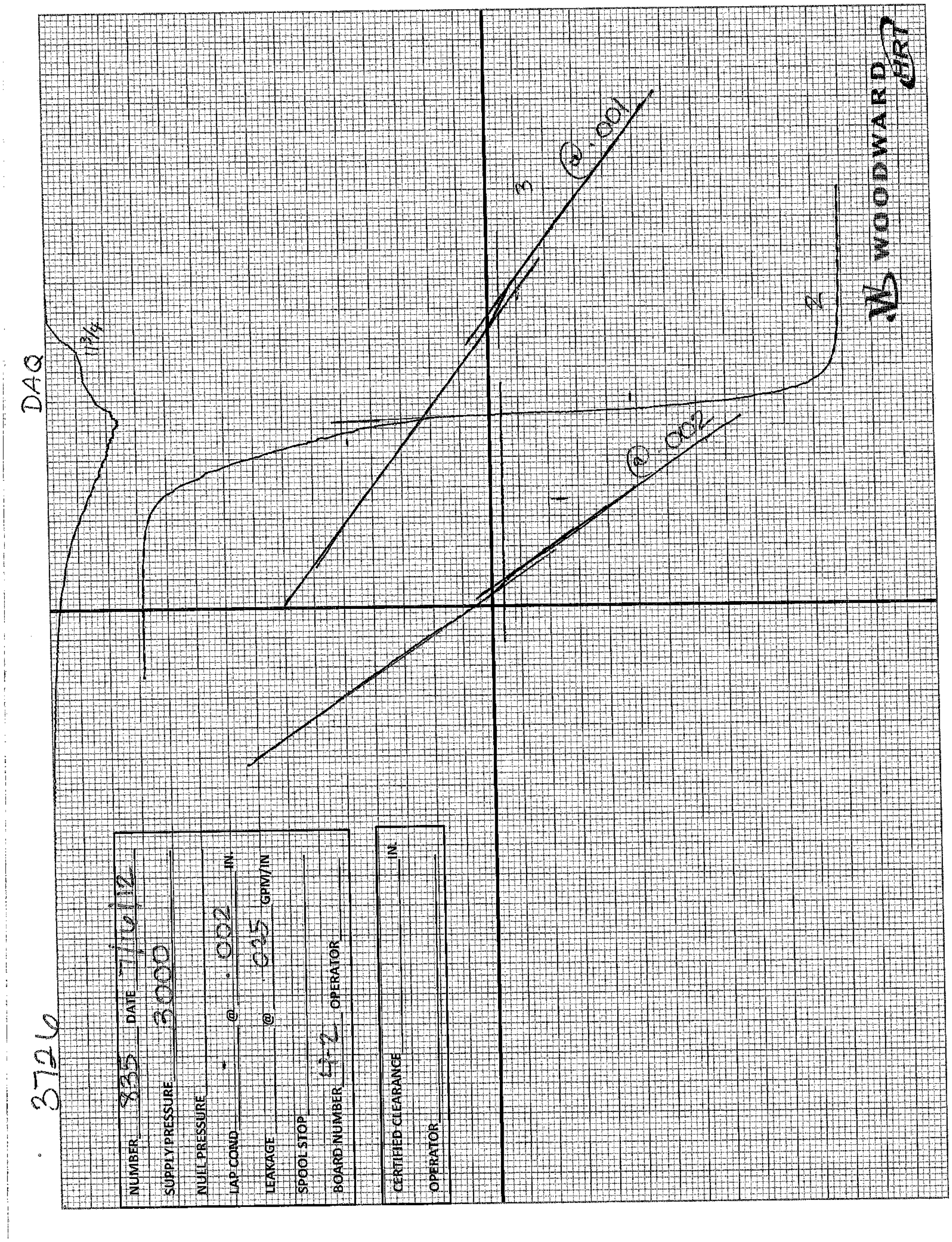
OPERATOR

WOODWARD HRT
25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355

DOCUMENT NO.: HR70600008



WOODWARD HRT

25200 West Rye Canyon Road 661/294-6000
Santa Clarita, California 91355